

Consensus guidelines for eligibility assessment of pathogenic variants to antisense oligonucleotide treatments: The N1C VARIANT guidelines

***Disclaimer**: This document is the work product of the N=1 Collaborative (the "N1C"). The N1C is not providing legal or regulatory advice for N=1 trials. This document should not be construed as legal or regulatory advice for any particular purpose. These guidelines focus on the evaluation of eligibility of a genetic variant for ASO treatment only. To assess a patient case for ASO amenability, the disease, phenotype, and patient have to also be considered, which is beyond the scope of these guidelines. The information is based on the best available knowledge and practice at the time of publication. Users of these documents should exercise their own judgment and discretion in applying these practices to their specific situations.*

The most up-to-date N1C VARIANT Guidelines and all accompanying training material and tools can be found on the N1C website: https://connections.n1collaborative.org/spaces/21600938/page

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General Overview & Introduction

These consensus guidelines are part of a set of guidelines and frameworks developed via the N=1 Collaborative (N1C) to enable clinicians and researchers within the rare disease community to assess and prioritize patient cases for genetic therapy development. The guidelines are being developed by the Patient Identification Working Group (PIWG), one of the N1C's working groups that aims at developing and disseminating guidance on identifying patients that may be amenable to experimental individualized genetic therapy intervention. The N1C VARIANT guidelines specifically focus on antisense oligonucleotides (ASOs) and siRNA, which are currently most often applied clinically. These guidelines are the first consensus guidelines that were developed and published through the N1C (Cheerie et al., 2025). Currently, these guidelines are most applicable for patients living with neurological disorders affecting the brain, spinal cord and eye, as well as liver diseases, given the current limited delivery capacities of ASOs to other target tissues.

These guidelines are solely focusing on the molecular assessment of pathogenic DNA variants. For better case assessments, these guidelines should be used alongside our framework on evaluating genes and diseases for genetic interventions (pre-print linked here).
Within the next year, we will further release guidelines for assessments of variants towards gene editing and gene replacement therapies to complement our efforts on streamlining N=1 therapy developments.

Consensus guidelines for eligibility assessment of pathogenic variants to antisense oligonucleotide treatments: The N1C VARIANT guidelines

[v2.0 July 2026]

(written by David Cheerie and Marlen Lauffer together with Emma Sherrill, Logan Newton, and Aubrie Soucy, on behalf of the N1C Patient Identification Working Group (PIWG)). Original publication: 10.1016/j.ajhg.2025.02.017

Purpose

The N1C VARIANT (**V**ariant **A**ssessments towa**r**ds El**i**gibility for **An**tisense Oligonucleotide **T**reatment) guidelines provide professionals working in the rare genetic disease field (e.g., clinical geneticists, diagnostic laboratories, rare disease researchers) with a framework for analyzing and classifying disease-causing variants for their amenability towards antisense oligonucleotide (ASO) therapies. Specifically, these guidelines are meant to identify genetic variants most likely to benefit from ASO-based therapy and distinguish these variants from currently less suitable candidates. With this, the guidelines address the very first step in the assessment of an individual toward ASO development (see graphical abstract). Using these guidelines, assessors should be able to:

1. Identify pathogenic variants eligible for analysis by these guidelines and utilize publicly available databases and resources to assist in the variant analysis process
2. Identify pathogenic variants causing aberrant splicing and assess what type of splice-altering variants can be targeted using ASOs
3. Assess whether a pathogenic variant is eligible for treatment by exon skipping ASOs
4. Assess whether candidate genes and/or variants are eligible for siRNA or ASO-mediated transcript knockdown
5. Classify variants as either “eligible”, “likely eligible”, “unlikely eligible”, “not eligible”, or “unable to assess” towards each of splice correction, exon skipping, or transcript knockdown approaches. The definition of each classified variant is dependent on the type of RNA therapy and is further described in each respective section
6. Consider strategies for upregulation of wildtype alleles in cases of haploinsufficiency

To follow these guidelines, readers should have an understanding of foundational genetic concepts including, but not limited to, splicing, introns and exons, coding versus non-coding, and DNA variant types (nonsense/stop-gain, missense, indels, frameshifts, etc.). While these guidelines will remind assessors of key definitions (canonical splice site, cryptic splice site, etc.), these concepts will not be explained in detail. Assessors should ideally also be familiar with standard variant annotations and pathogenicity classification approaches, including preferably the ACMG-AMP guidelines (Richards et al., 2015).

These guidelines focus on the evaluation of eligibility of a pathogenic genetic variant for ASO treatment. To assess a person's eligibility for ASO treatment, disease- and individual-specific clinical factors have to also be taken into account, which is beyond the scope of these guidelines (Lauffer, van Roon-Mom, Aartsma-Rus, & N = 1 Collaborative, 2024). Since these guidelines focus on assessing single pathogenic variants, they cannot be applied to marketed drugs and for more generalized assessment of therapeutic eligibility of a gene.

Please also consider that an ASO therapy might not always be the most suitable option. Furthermore, in cases where no ASO can be developed, other therapeutic modalities should be considered. The PIWG is also developing guidelines for assessing variants towards gene editing and gene replacement approaches.

The guidelines were prepared to the best of our current knowledge and are subject to change, with new knowledge on the topic being generated continuously. These guidelines reflect a general way of evaluating variants, but there are many exceptions to this. Where necessary, we have mentioned relevant exceptions within the text or as footnotes. This also means a variant's classification can change over time and it might thus be useful to re-assess variants at a later stage. Classifications are made for a variant's eligibility towards a specific ASO strategy. For example, a variant might be classified as "not eligible" for splice correction, while also being classified as "likely eligible" for a transcript knockdown.

We recommend only assessing disease-causing variants classified as likely pathogenic and pathogenic according to the ACMG-AMP guidelines (Richards et al., 2015). An exception to this rule are VUSes (variants of unknown significance) that are *in trans* with a (likely) pathogenic variant in a recessive disorder.

For the assessments, all recommendations on suitable tools, websites, and databases to aid in the process are limited to publicly available resources, but other licensed resources can also be used at an individual's discretion. Instructions on how to use the recommended tools and websites are beyond the scope of these guidelines. Assessors are encouraged to familiarize themselves with the tools by utilizing the respective tools' "help" pages or corresponding research articles.

Structure of guidelines

These guidelines serve as an aid for assessing a given disease-causing variant and are not meant to be read as a whole. Instead, only specific sections need to be read for each assessment. The guideline will first provide the assessor with a background on ASO/siRNA therapies. The guidelines are then divided into 4 steps. Steps 0-2 are necessary to collect information, such as the inheritance pattern or pathomechanism of disease, relevant to each assessment and to decide on the required therapeutic approach. Step 3 is further relevant for each variant assessment and focuses on splicing evaluation, whereas Step 4 provides an overview (Fig. 7) to guide you to relevant sections of the guideline for specific assessments depending on the information gained in Steps 0-3. A quick overview of the structure can be seen in Fig. 4.

Sections A-C can then be read independently and the section matching the variant under assessment can be selected. Within the guidelines, assessors will be provided with the possibility to jump between sections via hyperlinks.

At the end of each section, we have formulated important considerations for each assessment. To support the understanding of the guidelines and their use, we have generated several example assessments and matching training videos.

In addition, the AVEC tool can support assessors in conducting variant assessments. It automates the collection of relevant data (e.g., inheritance, dosage sensitivity) while allowing users to input or override information as needed. Using this data, the tool applies the N1C VARIANT framework to classify variants across different ASO strategies, and provides both the final classification and the underlying rationale for transparency and interpretability.

Guideline in Brief

Pathogenic DNA variants can be assessed for ASO eligibility using these guidelines. ASO approaches to be considered during the assessment are splice correction, exon skipping and inclusion, transcript downregulation, and upregulation of the wildtype transcript.

Variants that can be considered for evaluation with these guidelines are:

- Diagnostic DNA variant (disease-causative pathogenic or likely pathogenic variant)
- Non-mosaic
- Non-mitochondrial

AND one of the following…

- Single-nucleotide variant
- Small insertion/deletion
- One or multi-exon deletion/duplication
- Single-gene deletion/duplication

These guidelines do not apply to: variants in non-coding genes (some exceptions apply), deletions/duplications spanning multiple genes, imprinting defects/uniparental disomy, structural rearrangements (e.g., translocations), and aneuploidies.

Assessment of (likely) pathogenic DNA variants for eligibility towards ASO approaches, can be divided into multiple steps. This includes determining 1) the inheritance pattern of the variant, 2) the pathomechanism of the variant and dosage sensitivity of the gene, and 3) the splicing effect of the variant. Further, each variant and – where applicable – disease should be checked for already existing therapies before an individualized therapy should be initiated.

With the relevant information in place, suitable ASO approaches can be identified and the respective section of the guideline used for assessment (Figure 4 and 7).

Splice Correction: Can be considered for splice-altering variants independent of inheritance pattern and pathomechanism. Caution should be paid to pathogenic variants disrupting the canonical splice sites or branch points as these are usually not suitable candidates for splice correction.

Exon skipping: Exon skipping can be considered for all variants independent of inheritance pattern or pathomechanism but is most suitable for loss-of-function variants in dominant and recessive disorders. Exon skipping assessments are most complicated, and only a small subset of variants fulfils the eligibility criteria.
Transcript knockdown: Knockdown of mutant transcripts is most suitable for gain-of-function and dominant negative variants in dominant disorders. It should be distinguished between allele-selective and non-allele-selective approaches, where the approach is determined by the dosage sensitivity of the gene. For dominant negative variants, only allele-selective approaches are applicable.
Upregulation from the wildtype allele: Generally, to be considered for haploinsufficiency disorders, that is loss-of-function variants in autosomal dominant disorders.

Each assessed variant is classified towards a different approach as “eligible”, “likely eligible”, “unlikely eligible”, “not eligible” or “unable to assess” to indicate molecular eligibility and ultimately to prioritize candidates for further consideration for ASO developments.

Importantly, a molecularly targetable variant does not necessarily translate into a strong clinical indication. Priority is highest for well-characterized monogenic diseases with a defined natural history, a feasible therapeutic window, and no effective existing treatments. Delivery remains a key constraint, as efficacy depends on reaching the target tissue and achieving sufficient editing. Conditions where partial rescue improves quality of life, and those with measurable biomarkers, are particularly valuable for early translation. Further clinical considerations are outlined in N1C’s “Consensus framework to assess eligibility of genes and hereditary diseases for genetic therapy development.”

Background

ASOs are short, synthetic, single-stranded oligonucleotides that can bind RNA via Watson-Crick base pairing. ASOs can modify protein expression through various mechanisms (Dhuri et al., 2020, Rinaldi & Wood, 2017).

Splice Modulation

The binding of ASOs to splice sites or splice-regulatory elements on the pre-mRNA transcript allows for manipulation of the splicing process, which can lead to (canonical) exon skipping/inclusion or the restoration of wildtype splicing.

ASOs can be used to restore wildtype splicing in individuals whose mechanism of pathogenicity is caused by the activation or creation of cryptic (non-canonical) splice sites, resulting in aberrant splicing of the transcript (Fig. 1A) and, for example, the inclusion of parts of the intron then termed cryptic or pseudoexon. Additionally, canonical exon skipping ASOs can be used to “skip” exons containing the pathogenic variant, to produce a truncated, yet functional, protein product (Fig. 1B) in the case of loss-of-function (LoF) variants. For gain-of-function (GoF) variants, exon skipping can also be applicable, but there is no requirement to generate a functional protein product.

In rare circumstances, certain variants, like single nucleotide polymorphisms (SNPs) or pathogenic variants, lead to skipping of an exon and subsequently decreased production of a protein product. It is possible to develop ASOs that will lead to exon inclusion, however, the development of such efforts is challenging (Singh, Lee, DiDonato, & Singh, 2015) (Fig. 1C).

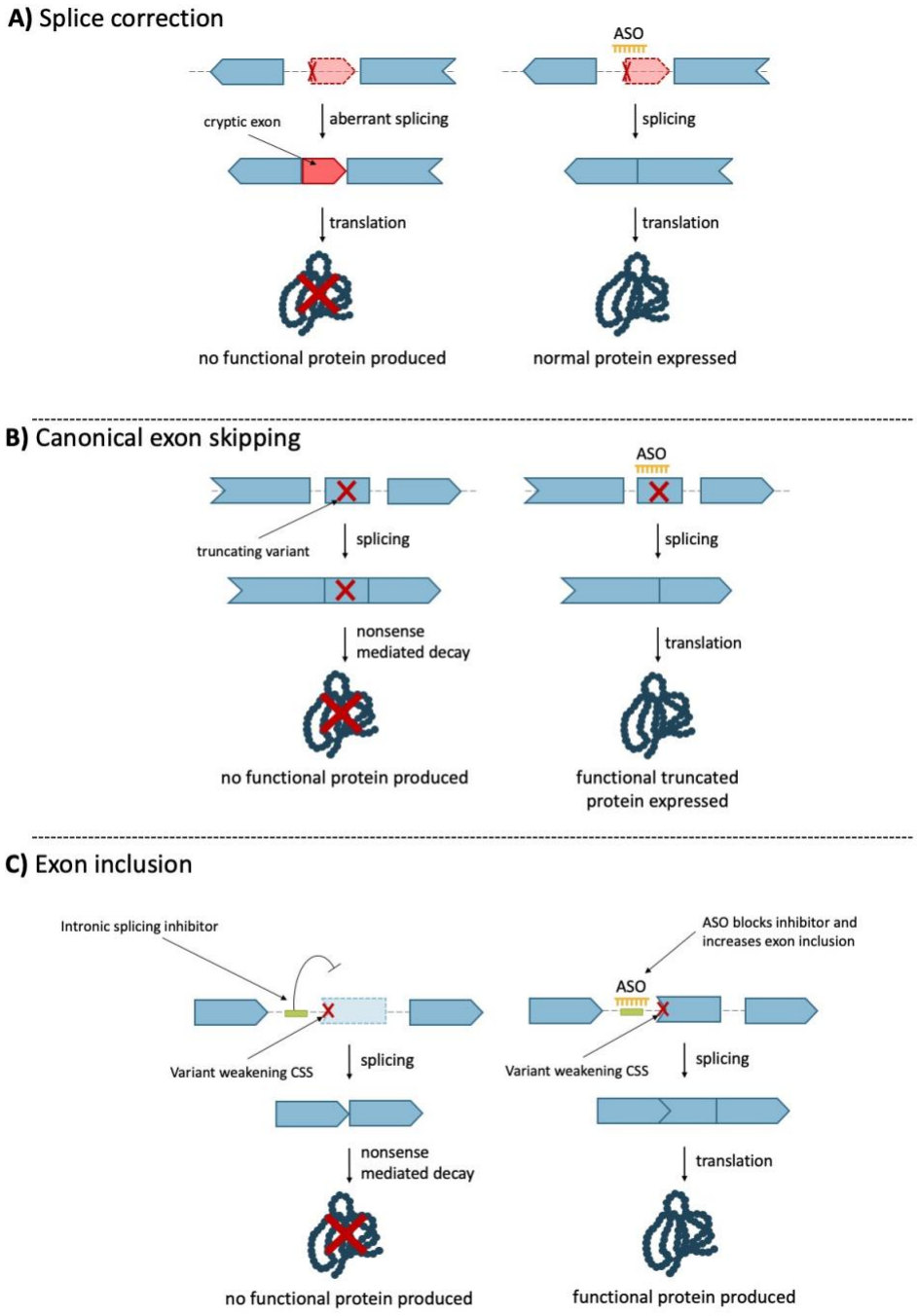


Figure 1: Splice modulating mechanisms of ASOs for the restoration of a functional gene product.
A) ASOs can be used to skip a cryptic exon caused by deep-intronic pathogenic variants to restore canonical splicing. B) ASOs can be used to skip an in-frame canonical exon containing a truncating variant to produce a truncated yet functional gene product. C) ASOs can be used to upregulate inclusion of an exon by blocking inhibitory sites.

Transcript Knockdown

In addition to splice modulation, oligonucleotides can bind to the target transcript and downregulate (pre-)mRNA expression (i.e., knockdown ASOs). Knockdown can be achieved with gapmer ASOs and small interfering RNAs (siRNAs). Gapmer ASOs trigger RNase H-mediated cleavage (Fig. 2), while siRNAs trigger the endogenous RNA interference pathway. Both mechanisms can be utilized in situations where the pathomechanism is a result of overexpression, toxic GoF, or dominant-negative (DN) effects (Lauffer, van Roon-Mom, Aartsma-Rus, & N = 1 Collaborative, 2024).

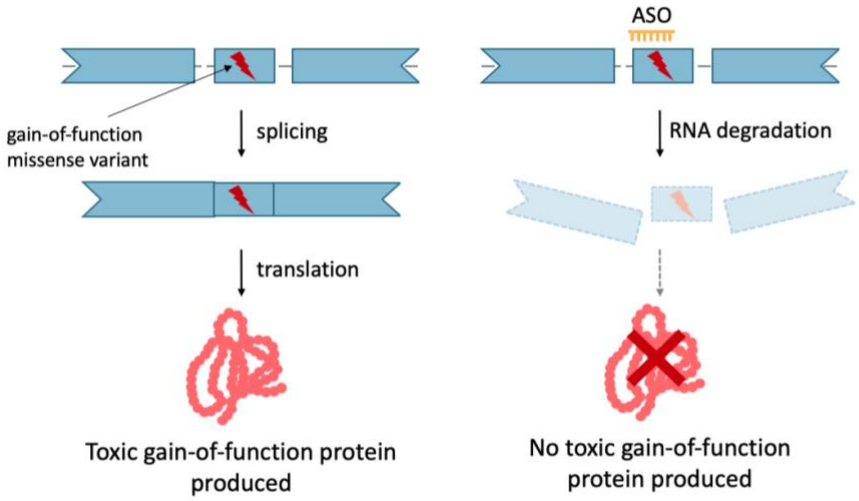


Figure 2: RNA knockdown using ASOs.

ASOs can be used to knockdown RNA transcripts which carry variants leading to a toxic GoF protein (shown here), proteins with a DN effect, or overexpressed proteins.

Increased protein from wildtype transcript

Furthermore, one can consider upregulation from the WT allele, such as targeted augmentation of nuclear gene output (TANGO), as described in Lim et al., 2020, Mittal et al., 2022, Felker et al., 2023, and Liu et al., 2022 (Fig. 3). For disorders caused by haploinsufficiency, one wildtype allele remains intact and functional. ASOs can be utilized to upregulate the gene product from the wildtype allele with the goal of restoring proper gene and cell function.

One such approach includes the targeting of poison exons. Poison exons are naturally occurring, highly conserved alternatively spliced exons that result in premature termination when included in the transcript. An ASO can be designed to skip poison exons and increase the number of productive/protein-coding transcripts, with the goal of increasing protein levels (Fig. 3A).

Additionally, ASOs can be designed to target naturally occurring antisense transcripts which are non-coding RNAs that can act on one or more corresponding transcripts with diverse roles, including RNA interference and RNA masking (Khorkova et al., 2022). In such cases, targeting antisense transcripts using ASOs can upregulate transcript levels (Fig. 3B).

Lastly, ASOs can be designed to target untranslated regions (UTRs) to upregulate or stabilize productive transcripts (Liang et al., 2017; Sasaki et al., 2019). One possible method is the targeting of the upstream open reading frames (uORF). These are alternative reading frames that occur upstream (5') from the canonical reading frame (primary ORF, pORF). These reading frames may code for proteins, but can also downregulate the reading of the canonical reading frame. ASOs targeting the uORF can be used to upregulate transcripts from the canonical reading frame. This can either be done by blocking the uORF or by skipping the exon containing the uORF (Fig. 3 and Fig. 11 in Section C). Another related method is the targeting of the 3' UTR by sterically blocking some inhibitory functional sites, such as microRNA binding sites. By using ASOs to interfere with degrading complexes, one can attempt to increase RNA half-life and increase gene product.

Overall, the upregulation of wildtype gene products is a possible approach for disorders caused by haploinsufficiency.

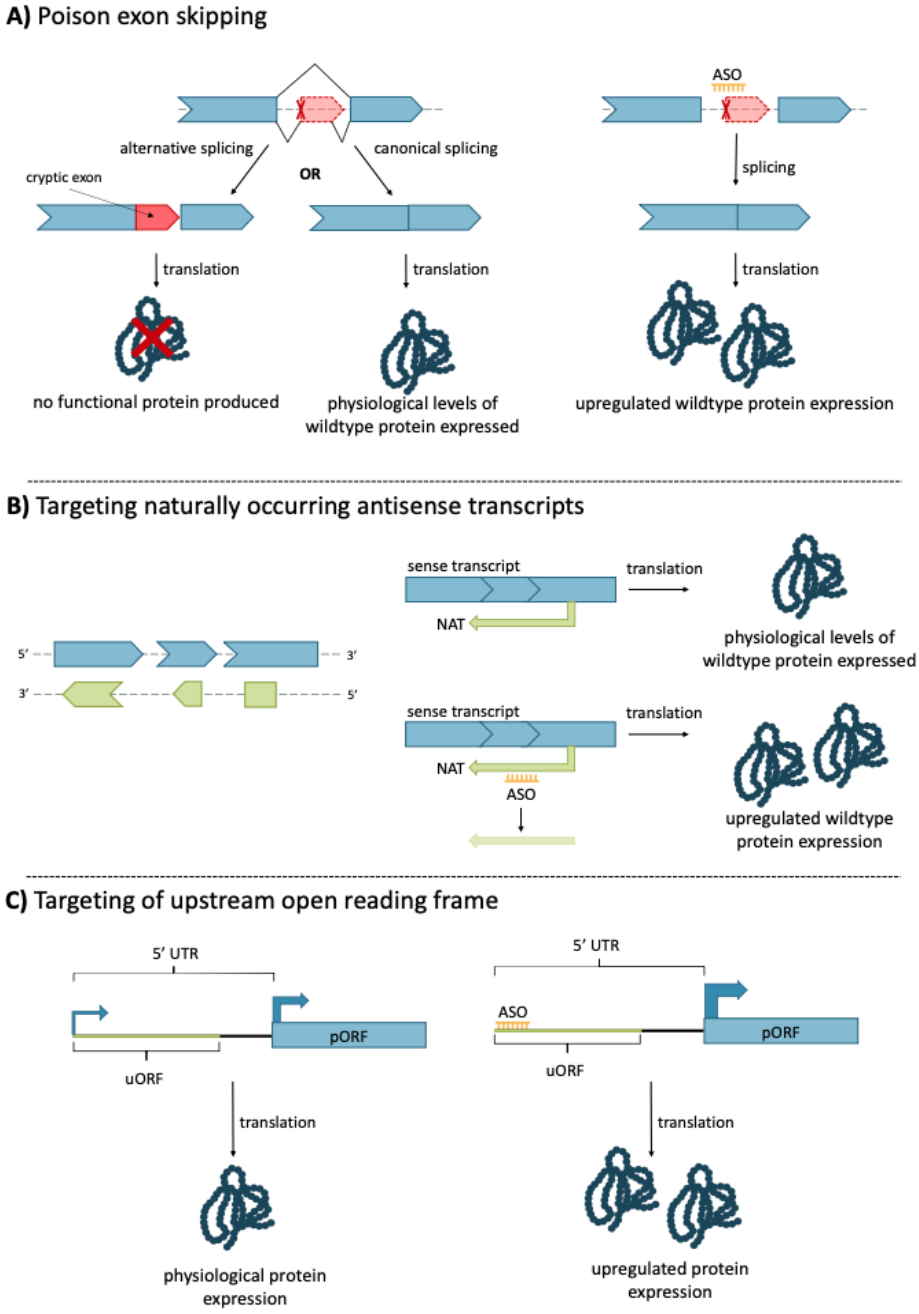


Figure 3: Upregulation of wildtype allele transcript using ASOs.
A) An ASO can be used to skip a poison exon in a transcript that would usually lead to nonsense-mediated decay, increasing wildtype transcript levels and subsequent gene product. **B)** ASOs can target NATs which negatively impact transcription of the sense transcript. By disrupting NAT transcription, the wildtype transcript can be increased. **C)** Targeting uORFs with an ASO can be used to promote translation of the pORF (primary open reading frame), increasing wildtype gene product.

ASOs offer a potential avenue for the treatment of rare genetic diseases, although this is currently mainly focused on disorders impacting the central nervous system (brain, spinal cord, retina), or liver, due to restricted delivery options. FDA- and EMA-approved ASO therapies include, for example, nusinersen for spinal muscular atrophy (SMA1 [MIM 253300]) (Egli & Manoharan, 2023). The use of ASOs to develop variant-specific therapies has, for example, been demonstrated through milasen, an ASO developed at Boston Children's Hospital, to target a deep intronic *MFSD8* [MIM 611124] variant resulting in aberrant splicing (Kim et al., 2019).

To better understand the concepts and guidelines shared in this document, it is important to familiarize yourself with the key terms outlined in Table 1.

Table 1: List of Key Terms

Term	Definition
Branchpoint	*cis*-acting intronic motif (specific intronic sequence on the same chromosome) required for pre-mRNA splicing, usually an A (adenosine) located 18-40 nucleotides upstream of the acceptor splice site.
Canonical Splice Acceptor Site	RNA sequence recognized by the spliceosome flanking the 3' end of an intron, usually an AU sequence.
Canonical Splice Donor Site	RNA sequence recognized by the spliceosome flanking the 5' end of an intron, usually a GU sequence.
Canonical Splicing	Splicing involving the use of the canonical acceptor and donor sites (see definitions for canonical acceptor site and canonical donor site).
Cryptic Splicing	Cryptic splice sites are naturally occurring splice sites within the genome that are used infrequently. Splicing involving cryptic splice sites, i.e., cryptic splicing, often leads to the incorporation of parts of an intron into the mRNA transcript (cryptic exon or pseudoexon) or the removal of parts of an exon ultimately leading to an early translation stop. Pathogenic variants causing aberrant splicing can for example activate, strengthen, or create a cryptic splice site and thus cause disease.
DNA Tandem Repeats	Short lengths of DNA repeated multiple times within a gene, e.g. CAG repeats in PolyQ disorders.
Hypomorphic Allele	Alleles that show partial loss of function. Sometimes referred to as "leaky" alleles because there is some retention of protein function.
In-Frame Exon	Exon in which the number of base pairs is divisible by 3. Since each amino acid is encoded in one codon made up of three base pairs, removing an exon that is a multiple of 3 will not disrupt the reading frame. While an exon can be divisible by 3, it does not mean that it starts with the first nucleotide of a codon and ends with the last nucleotide of a codon. The

	exon boundaries and codon boundaries do not necessarily align. See the "Assessing Exon Position and Frame" section in this document.
MANE Select	Matched Annotation from NCBI and EMBL-EBI (MANE): "The MANE Select set consists of one transcript at each protein-coding locus across the genome that is representative of biology at that locus. This set is useful as a universal standard for clinical reporting, as a default for display on browsers and key genomic resources, and as a starting point for comparative or evolutionary genomics. MANE Select transcripts are identified using computational methods complemented by manual review and discussion.'' (Morales et al., 2022).
Naturally Occurring Antisense Transcript	Noncoding antisense transcripts that can act on one or more corresponding transcripts with diverse roles, including RNA interference and RNA masking.
Out-of-frame Exon	Exon in which the number of base pairs is not divisible by 3. Please see "in-frame exon" for further explanation on exon frames.
Poison Exon	Naturally occurring, highly conserved alternatively spliced exons which result in premature termination when included in the transcript, either as a result of frameshift or the inclusion of a premature stop codon
Protein Tandem Repeat Domain	Two or more domains from the same family found in tandem, e.g. spectrin-like repeats in the dystrophin protein
Splicing	A step in the processing of mature mRNA in which introns (non-coding sequences) are removed or "spliced out" of the pre-mRNA transcript, and the remaining exons (coding sequences) are connected to one another forming the mRNA.
Splicing Enhancer Site (SE)	RNA sequence motif found in the exon/intron of genes, binds proteins that help recruit splicing machinery to the correct site, directing or enhancing accurate splicing.
Splicing Silencer Site (SS)	RNA sequence motif found in the exon/intron of genes, binds proteins that negatively affect the core splicing machinery, inhibiting or silencing the inclusion of an exon into the mRNA.
Upstream Open Reading Frame	Alternative reading frames that occur upstream (5') from the canonical reading frame.

Further Resources

For further resources on the mechanisms and use of ASOs for genetic disorders, please see the below recommendations. Note: these additional resources are not required but are encouraged for those who are not familiar with ASO technology.

Educational videos

Treating Disease at the RNA Level with Oligonucleotides:
https://www.youtube.com/watch?v=nRHypCupg0A

Modifying RNA splicing with Morpholino Oligos by Gene Tools:
https://www.youtube.com/watch?v=qu-Kz0HaLxw&ab_channel=GeneTools

Overview articles

Lauffer et al., *Possibilities and limitations of antisense oligonucleotide therapies for the treatment of monogenic disorders* (10.1038/s43856-023-00419-1) Communications Medicine

Hammond et al., *Delivery of oligonucleotide-based therapeutics: challenges and opportunities* (10.15252/emmm.202013243) EMBO Molecular Medicine

Variant Assessment

The variant assessment is divided into different steps (please also see flow diagram Fig. 4).
Step 0 - Variant check
Step 1 - Assessment of pattern of inheritance and disease type
Step 2 - Assessment of pathomechanism of the genetic variant and dosage sensitivity
Step 3 - Evaluation of splicing effects
Step 4 - Identification of relevant guideline
    Section A - Considerations for Canonical Exon Skipping
    Section B - Considerations for Transcript Knockdown
    Section C - Considerations for Upregulation from the Wildtype Allele

For the assessment of a variant for ASO eligibility, different types of information need to be gathered to decide on suitable ASO strategies and assess a variant using the specific sections (sub-guidelines). To determine the most suitable guideline for each variant, it is first necessary to check whether the variant can generally be assessed with these guidelines and whether the variant description is correct (Step 0), what the inheritance pattern of the variant and the disorders the gene is associated with are (Step 1), what the pathomechanism of the variant is (Step 2), and whether a variant is influencing splicing (Step 3). With this information at hand, the assessor can decide on possible ASO strategies and read up on the specific sections of the guidelines (Step 4) that will focus on strategies like exon skipping and knockdown approaches.

These guidelines are not meant to be read as a whole, but using the information gathered in Steps 1-3, the flow diagram in Fig. 4, and the detailed diagram in Step 4 (Fig. 7) will guide the assessor to the relevant sections. An exception is Step 3 which not only checks for influence on splicing but in case of effects on splicing already allows the assessor to classify the variant with respect to splice correction ASOs.

We further provide some notable exceptions and special cases as footnotes. These are cases that apply rarely but were added for the assessors to gain a comprehensive understanding of the assessment process.

Variants will be classified as “eligible”, “likely eligible”, “unlikely eligible”, “not eligible”, or “unable to assess” towards a specific approach (Table 2) using these guidelines with the exception of upregulation from the wildtype allele (Section C), where no such classification is possible.

Table 2: Explanation of Variant Classification Terms

Classification	Explanation
Eligible	Variants are considered eligible when functional evidence supports the effectiveness of an ASO approach. What type of functional evidence is deemed sufficient depends on the approach and is defined within the different sections below.

Likely eligible	Variants are considered likely eligible variants when the variant could potentially be targeted by an ASO, although no functional evidence is currently available to confirm this. That means a variant meets all criteria relevant for ASO development on paper.
Unlikely eligible	Variants are considered unlikely eligible variants when the molecular criteria suggest an ASO is, with our current understanding, unlikely to be effective, but no functional evidence directly contradicts the potential use of an ASO.
Not eligible	Variants are considered not eligible when an ASO approach will not work. This can be the case if the genetics do not allow for an ASO correction or an ASO cannot be designed. It could also be that there is evidence that demonstrates an ASO approach will not work. For example, exon skipping leads to a non-functional protein.
Unable to assess	All variants that currently cannot be assessed with these guidelines. This can be due to the variant not being applicable to these guidelines or that not enough information is available on a variant that allows for it to be assessed.

Please note that variants can be applicable and assessed for different approaches and while a variant might be “unlikely eligible” for splice correction, it could for example still be “(likely) eligible” for exon skipping. **Assessments may be overridden if the assessor provides appropriate justification and are not required to strictly adhere to the criteria outlined in the classification tables.**

The guidelines are also provided with a set of example variant assessments found at the end of this document. Videos accompanying the example assessments walk the assessors through the assessments to aid with training. Assessors can further use the test variants that were assessed during the consenting process for further practice. All the original test variants used during the development of version 1.0 can be found in Table S1, and corresponding answers in File S2 of the original publication (https://doi.org/10.1016/j.ajhg.2025.02.017). New or updated test variants released alongside later versions of the guideline can be found here: and on the N1C website. The N1C Gene Registry further contains a section displaying variant assessments and their outcome and can be used to practice assessments (https://generegistry.n1collaborative.org/assessed_variants.html).

Together with v2.0 of the guidelines, we have developed the “Advanced Variant Eligibility Calculator” (AVEC, https://avec.uni-koeln.de/) to support and accelerate assessments. This tool replaces the N1C Variant Eligibility Calculator developed for v1.0 of the guidelines and is a significant improvement as it semi-automates part of the assessment and provides relevant information in one interface. The AVEC tool also allows for commenting on the assessment and downloading a report.

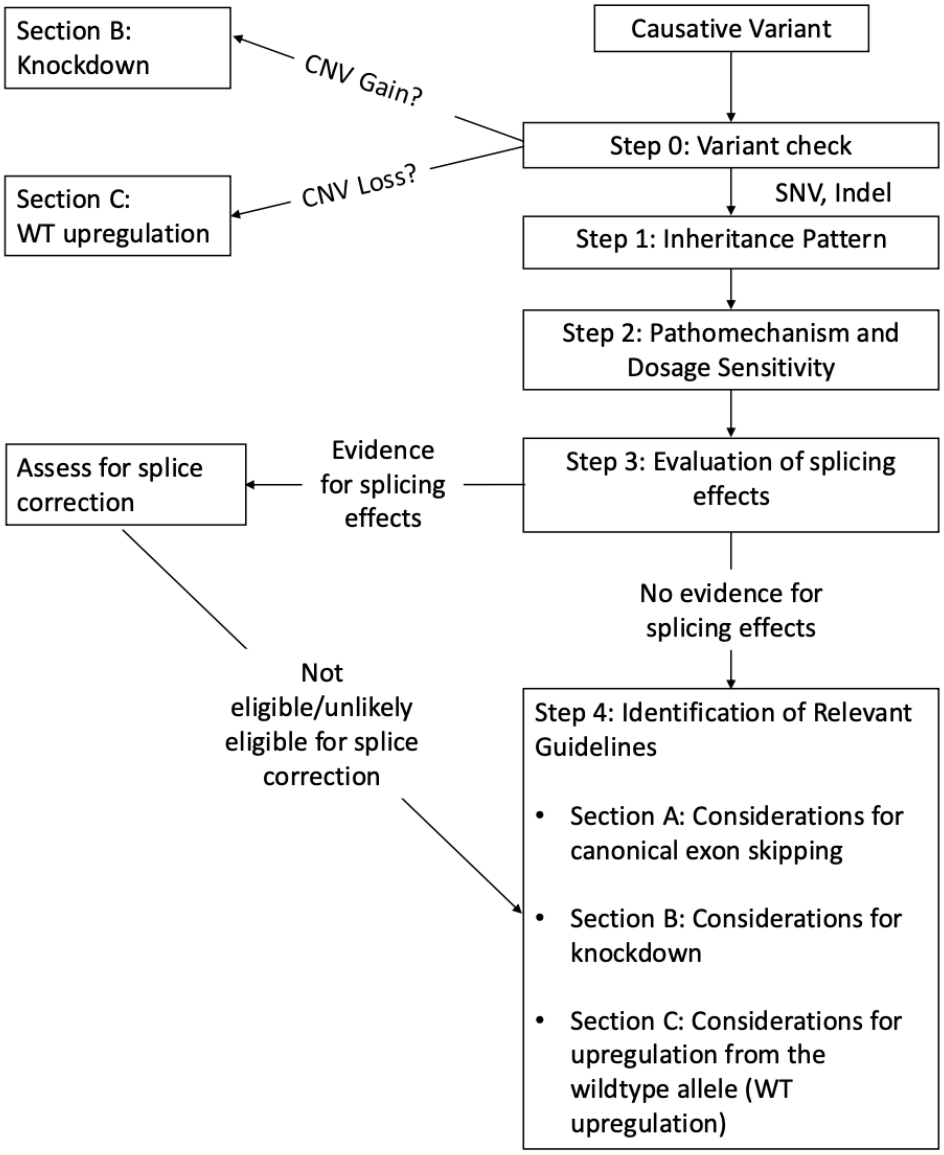


Figure 4: Overview of consensus guidelines document.
The guidelines begin with Step 0, 1, and 2, which highlight the assessment of the variant description, inheritance pattern, and pathomechanisms of the variant. When applicable, the evaluation of splicing effects is considered in Step 3. Depending on the information gathered in Steps 0-3, readers are directed to Sections A, B, or C. The document is designed in a way in which assessors are directed to relevant sections, and are not required to read the entire document for assessment of a variant.

Step 0 - Variant check

In this step, the assessor will learn which type of genetic variant can be assessed using these guidelines, and how to ensure that the variant description provided is correct. Without the correct variant description, an assessment is not possible.

Variants applicable to these guidelines are restricted to (likely) pathogenic, disease-causing single nucleotide variants, small indels, one or multi-exon deletions or duplications, and single gene deletions or duplications. Currently excluded are variants in non-coding genes[1], deletions/duplications spanning multiple genes (i.e., contiguous gene syndromes), imprinting defects/uniparental disomy, structural rearrangements (e.g., translocations), and aneuploidies. Additionally, mitochondrial DNA disorders (i.e., variants in the mitochondrial genome) cannot be evaluated with these guidelines. Furthermore, due to challenges associated with the delivery of ASOs and targeting of mosaic variants, these variants are not currently eligible for assessment by these guidelines. All variants that cannot be evaluated will be classified as “unable to assess”. Insertions within coding regions are mostly not eligible unless confined within an exon that can be skipped; insertions in introns can be applicable, one example being milasen (Kim et al., 2019). Repeat expansion disorders can also be assessed using these guidelines but the pathomechanism should be well understood to evaluate for knockdown or exon skipping approaches.

Before starting to assess a variant, it is important to check the accuracy of the variant description. That means, is the gene symbol correct, does the transcript match the variant and is the consequence on protein level denoted accurately? If the variant description is incorrect, the variant should be classified as “unable to assess” until the description is corrected and verified. These guidelines work with the MANE select transcript but can also apply to any other transcript. Choosing a different transcript when it has more biological relevance (e.g. MANE clinical) in a given disease context is preferred.[2]

To check a variant and its description, please follow the HGVS nomenclature. Correct variant description and matching of gene and transcript can be checked with Mutalyzer and VariantValidator. Examples of variants and their descriptions are provided at the end of these guidelines.

Considerations for Single Gene Copy Number Variants (CNVs)
For the gain or loss of whole genes, special considerations apply. For the gain of a whole gene, where dose amplification is the pathomechanism, knockdown strategies should be considered (see Section B). For the loss of a gene with one functional wildtype copy left, upregulation from the WT allele should be considered (see Section C). ASO strategies cannot be applied when no

[1] Some exceptions for non-coding genes can be made when a clear pathomechanism of disease is defined. For example, for RNU4-2 snRNA a knockdown ASOs can be used therapeutically.
[2] In some instances, variants that are deep intronic in the MANE select transcript are exonic in another transcript and are disease-causing due to the effect they have in that exon. Please pay attention to this exception when evaluating the functional consequences of a variant.

copies of the gene are present, i.e., loss of both gene copies or loss of one gene copy for hemizygous genes.

Optional ASO Check After a Variant Check

To save time for assessors familiar with these guidelines, we recommend checking whether there has already been an ASO/siRNA developed for the variant in question (clinically or pre-clinically) immediately after checking the correct variant description. This can mean that an ASO has been developed for the specific variant, e.g., a splice correction ASO or a gapmer ASO, or an ASO has been developed for an exon skipping approach for an exon this variant is located in. It can also be that a gapmer ASO/siRNA is available for the gene in question or allele-selective for a SNP that is in phase with the pathogenic variant. When doing this ASO-availability check, it should be ensured that the ASO strategy identified also applies to the variant under assessment. If in doubt, we strongly suggest following the full guidelines and checking for available ASOs at the end. Generally, if an ASO has been developed, it should be carefully evaluated whether there is enough functional evidence that the ASO development was successful. That means, for example, demonstrating restoration of protein levels or rescue of a cellular phenotype. In cases where there is enough functional evidence, the variant can be classified as “eligible”, and no further evaluation is necessary[3,4]. To investigate whether an ASO has already been developed, we recommend a thorough search using ClinVar, Pubmed, Google Scholar, and web search, also paying attention to conference abstracts if available. Further, different ASO registries are available. For example, the N1C Gene Registry provides an overview of developed and ongoing N=1 projects, marketed genetic therapies, and also already assessed variants. The n-Lorem foundation also has an overview of their active gene programs available. We consider it sufficient when there is an ASO/siRNA already in clinical implementation, even if there is not yet published data available. That means if an ASO is available in the registries and preclinically validated or has already been administered to one or more individuals, this is sufficient to classify the variant under assessment as “eligible”.

We also encourage assessors to conduct a general review of the literature and clinical trial databases to identify other genetic therapy approaches under investigation. In doing so, it is important to pay attention to the specific indication and the underlying pathomechanism targeted by these therapies, as these can differ between variants within the same gene. In some cases, alternative therapeutic strategies may already be in development or have demonstrated efficacy, making an ASO approach unnecessary.

[3] Please note that if publications are available that test ASOs for a specific variant/exon/gene, one has to carefully read whether the ASO design and development was indeed successful. It could be that ASOs were designed but the transcript or protein level could not be restored. Now it depends whether it might not be possible at all to, for example, skip a canonical exon, or whether another ASO design approach might still be justifiable. Depending on the assessment, that might lead to a “not” for eligibility or a “likely/unlikely” following the guidelines. We generally recommend that at least two independent groups with sufficient functional evidence or one group with clear explanation of mechanism should have shown that ASO development is not possible to declare a variant “not eligible”.

[4] In the case of an allele-selective ASO development, please check whether the ASO was developed for the specific variant or for a SNP. For the latter, only if the individual has that exact SNP in phase with the pathogenic variant would that ASO be applicable. If this is not the case, proceed with the next steps of the assessment.

Step 1 - Assessment of pattern of inheritance and disease type

In this step, assessors will identify the pattern of inheritance of the variant under assessment and the disorders implicated in the disease gene. Understanding the inheritance pattern of the variant and the diseases associated with a gene is crucial to later decide on the most suitable ASO strategy and thus the section to use for the assessment. Different considerations will apply depending on the inheritance pattern.

The assessor has to identify if the variant is inherited in an autosomal dominant (AD), autosomal recessive (AR), X-linked, or Y-linked manner. Usually, in the case of (autosomal) recessive inheritance, the variant is either being reported as homozygous or a second variant *in trans* has been identified (i.e., compound heterozygous), making this step of the assessment straightforward. In the case of X-linked disorders in XY males, the variant will be hemizygous. Y-linked variants will also be hemizygous. For (autosomal) dominant inheritance, one variant should be reported either *de novo* or inherited from one of the parents. Should the respective information not be available (e.g., the phase of two variants in an autosomal recessive disease gene), some other steps can be taken to gather the necessary knowledge.
The following websites can be used to identify the inheritance pattern of a disease:
OMIM https://www.omim.org/
Orphanet https://www.orpha.net/consor/cgi-bin/index.php
GeneReviews https://www.ncbi.nlm.nih.gov/books/NBK1116/
Pubmed https://pubmed.ncbi.nlm.nih.gov/
ClinGen https://www.clinicalgenome.org/

Some genes are implicated in different diseases which can, for example, have an autosomal dominant and recessive pattern of inheritance respectively (e.g. *ACTL6B*: Developmental and epileptic encephalopathy 76 (OMIM 61846, autosomal recessive), Intellectual developmental disorder with severe speech and ambulation defects (OMIM 618470, autosomal dominant)). In such instances, it is important to identify which pattern of inheritance applies to the specific case. A web search on the variant can be useful. The variant may have been reported in a publication where a pattern of inheritance is noted. Also, gnomAD can be of help. The population frequency of a variant can indicate if the variant is associated with dominant or recessive inheritance. For example, if a LoF variant has a high allele frequency for heterozygotes but no homozygotes are reported, the variant is more likely to be associated with an AR pattern of inheritance. Similarly, for X-linked disorders in XY males, a lack of hemizygotes would be the equivalent assessment. Further, checking with the diagnostic laboratory or the treating clinician and reviewing the family history can be useful to gather more information on the inheritance pattern in that specific case.

If a gene is implicated in multiple disorders with distinct inheritance patterns, we recommend documenting this, as it can inform downstream steps of the assessment. For example, a gene associated with an AD disorder caused by heterozygous GoF variants, and also with an AR disorder caused by LoF variants, suggests that a knockdown strategy targeting the GoF variant

is possible but that loss of too much of the gene product will also be detrimental (in some cases, knockdowns exceeding 50% can be detrimental, please see discussions on allele-selectivity and dosage sensitivity in Step 2).

Special considerations for X-linked disorders

Traditionally, diseases implicated by genes on the X chromosome can be inherited in either an X-linked recessive (XLR) or dominant (XLD) fashion. However, sex genetic inheritance can be complex, and these complexities must be taken into consideration.[5]

One mechanism to be taken into consideration when assessing ASO amenability is X-chromosome inactivation (XCI). XCI is the random process of silencing most of the genes on one copy of the X chromosome in XX females (Wu et al., 2014). That means that only one of the gene copies is getting expressed in each cell in females (i.e. only one X is active). In LoF heterozygotes, 50% of the cells express the healthy copy of the gene with normal function, while the other 50% express the mutant copy with no leftover gene function (Basava et al., 2025). If XCI is skewed, this ratio can deviate from 50/50, leading to significantly more or less cells with adequate gene expression. The skewage can also vary between tissues. In contrast, genes that escape XCI ("XCI escapees") are pseudo-autosomal (i.e. show biallelic expression). It is estimated that around 15% of genes in human fibroblast lines evade XCI (Wu et al., 2014). It is important to understand a disease gene's relationship with XCI to appropriately consider its eligibility for therapy development and the most suitable therapeutic approach. For example, disorders caused by variants in genes that escape XCI should be evaluated as autosomal disorders, whereas genes that undergo XCI require further consideration regarding dosage.

[5] Given the complexities of sex chromosome genetics, there has been some discussion in the field about expanding X-linked inheritance categories to account for disease penetrance, X-chromosome inactivation (XCI), and male lethality (Basava et al., 2025). Duchenne muscular dystrophy and Danon disease are examples where the traditional XLR/XLD classifications don't appropriately convey affected status (Basava et al., 2025).

Step 2 - Assessment of pathomechanism of the genetic variant and dosage sensitivity

In this step, assessors will identify the pathomechanism of the variant and assess whether the gene in question is associated with haploinsufficiency and/or triplosensitivity. The pathomechanism is relevant to decide on the most applicable ASO strategy and with this, the sub-guidelines to use for the assessment. Identifying whether a gene is associated with haploinsufficiency is crucial to deciding whether ASO approaches need to be allele-selective and whether the assessors should consider the guidelines on upregulation from the wildtype allele (see Section C). Further, triplosensitivity might limit the amount of upregulation suitable for a given gene.

A pathogenic variant can lead to different effects. The variant can lead to a loss of function of a protein, a toxic gain of function, or a dominant-negative (DN) effect. For an explanation of the different pathomechanisms, please see, e.g., Backwell & Marsh, 2022.

Assessing the variant effect can mostly be done by conducting a web search and reading up on publications and reports of the variant. Resources that can help in identifying the pathomechanism associated with a variant are:
OMIM https://www.omim.org/
Orphanet https://www.orpha.net/consor/cgi-bin/index.php
GeneReviews https://www.ncbi.nlm.nih.gov/books/NBK1116/
Pubmed https://pubmed.ncbi.nlm.nih.gov/
LitVar[2]: https://www.ncbi.nlm.nih.gov/research/litvar2/
ClinGen: https://clinicalgenome.org/
Decipher: https://www.deciphergenomics.org/

We recommend looking for functional studies that have tested the variants in relevant tissues and modelling systems. We do not recommend relying on prediction tools, as these will not provide the level of evidence required for assessing for therapeutic eligibility. The performed studies should have thorough and well-described methodologies, including relevant controls. For some variants, ClinVar now also reports functional evidence that can be taken into account for the assessment.

Note that for most variants, no functional studies have yet been performed, especially if the variants are very rare. However, there are some considerations that can help define the pathomechanism. For instance, it might be possible to infer the pathomechanism of a variant from the clinical presentation and response to certain drugs. Here, we recommend discussing with the treating clinicians and experts on the disease whether the variant can be considered GoF/LoF based on the clinical presentation.

Besides the clinical presentation, pathomechanisms can sometimes be inferred from the variant itself. We list some considerations on how to approach the pathomechanism of a given variant.

However, please note that these are assumptions and are not meant to replace the requirement of having functional evidence validating the pathomechanism:

- Nonsense and frameshift variants leading to an early protein truncation are usually predicated with a LoF effect as they often result in a null allele (Abou Tayoun 2018). However, multiple examples have emerged in the past years where nonsense variants were shown to cause aberrant splicing or early truncations leading to a stable protein with a DN effect (Sinha Ray et al., 2022, Levy et al., 2023). We thus do not recommend assuming LoF for nonsense and frameshift variants unless other indicators and knowledge on the gene mechanisms are well established.
- In a disease that has only been associated with LoF variants, a newly reported likely pathogenic or pathogenic missense variant may be LoF, especially if this variant is *in trans* with a known pathogenic LoF variant (for recessive diseases).[6]
- For genes where both LoF and GoF are a known cause of disease, assessing the pathomechanism of a missense variant is challenging. It is important to take into consideration the phenotype associated with the variant to make a decision on the pathomechanism. It can, for example, be possible to predict a pathomechanism in cases where LoF and GoF variants lead to distinguishable phenotypes. However, there are only rare cases where phenotypes are distinct enough to make such a decision. One such example includes distinctly different phenotypes associated with GoF and LoF variants in *GABRB2* [MIM 600232] individuals (Mohammadi et al., 2024). In cases where LoF and GoF variants lead to similar phenotypes as seen in intellectual disabilities, more evidence is necessary and functional studies are crucial. Generally, when in doubt, functional evidence should be obtained.
- The observed inheritance pattern can also be considered when determining variant pathomechanism. In some instances, where both AD and AR inheritance patterns are associated with disease, autosomal dominant variants are associated with a GoF or DN variant, while AR variants are associated with LoF variants. One such example includes *ACTA1* (Sparrow et al., 2003). Similar to using phenotypes to determine mechanism, functional evidence should be obtained when in doubt. There are rare reports of homozygous GoF (Drutman et al., 2017; Schwarz et al., 2020) variants, thus carefully checking the inheritance pattern and pathomechanism is important.
- DN variants are, by definition, dominant and can thus only be found in dominant disorders. However, distinguishing whether a variant is GoF or DN might be difficult. Fortunately, for GoF and DN variants, the ASO approaches are similar.
- A missense variant might cause the loss/disturbance of an inhibitory domain causing a toxic gain of function effect of the protein (Mohassel et al., 2021).

[6] We recommend restricting these assumptions to AR diseases because in some cases nonsense and frameshift variants can cause splice aberrations and ultimately lead to a GoF effect on protein level (Flanagan et al., 2017).

Regardless of the considerations above, if sufficient functional evidence does not exist for a given variant, the next step would be to request more information on the variant or experimentally determine the pathomechanism. In the meantime, classify the variant as “unable to assess”.

If sufficient functional evidence indicates that a variant exerts both GoF and LoF effects, it is important to delineate through functional studies which mechanism primarily contributes to the clinical phenotype and whether targeting that mechanism is likely to be therapeutically effective (e.g., will a knockdown strategy directed at the gain-of-function component meaningfully ameliorate the phenotype?). Additionally, approaches such as exon skipping and splice correction are not inherently restricted to either GoF or LoF mechanisms and may represent viable strategies depending on the context.

Haploinsufficiency

Besides the identification of the inheritance pattern and the pathomechanism of a variant, the associated gene also needs to be evaluated for haploinsufficiency. Haploinsufficiency refers to a situation in which one healthy, wildtype allele does not generate sufficient protein product to preserve the physiological state (Deutschbauer et al., 2005). In most cases, haploinsufficiency is connected to AD disorders associated with LoF variants; however, several genes have been identified that are associated with LoF variants causing haploinsufficiency, and also GoF variants. That is the case for example for the genes *SCN2A* [MIM 182390] and *SCN8A* [MIM 600702] (Li et al., 2021; Wagnon et al., 2017). Here, GoF and LoF variants cause different clinical presentations depending on the pathomechanism of the variant.

The knowledge of whether a gene is associated with haploinsufficiency is important when deciding on the best therapeutic approach for a variant (see Fig. 7). In brief, for AD disorders caused by LoF variants, in the case of haploinsufficiency, the healthy wildtype allele could be upregulated using an ASO strategy (see Section C). For GoF variants, it will be important to assess whether an allele-selective approach is necessary for transcript knockdown (see Section B and further information below).

While haploinsufficiency will influence the decision on the therapeutic strategy as indicated in Fig. 7, we recommend always assessing each variant for splice correction (Step 3) first as this is the most straightforward and most promising strategy. Only if splice correction is not a suitable option consider moving towards other approaches.

To determine whether a gene is associated with haploinsufficiency, different resources can be used. Often, a web search can help as well as GeneReviews.
GeneReviews https://www.ncbi.nlm.nih.gov/books/NBK1116/
Pubmed https://pubmed.ncbi.nlm.nih.gov/

Further indications of whether a gene is associated with haploinsufficiency can be gained from checking the gene constraint metrics like the pLI and LOEUF scores in gnomAD, or the haploinsufficiency score determined by the ClinGen consortium. It is also possible to refer to the pHaplo scores available via the UCSC Genome Browser (Collins et al., 2022). Additional

information on dosage sensitivity in general and how one can assess dosage sensitivity is provided in Section B. We consider a curation from a reputable independent source, such as the ClinGen consortium, the highest level of evidence for dosage sensitivity/haploinsufficiency.

Triplosensitivity

Triplosensitivity is referred to a stage in which an organism is intolerant to a duplication of a gene, ending up with three copies. Upregulation therapies could in theory increase the gene product above the physiological level, negatively impacting the physiological state. To identify whether a gene is associated with triplosensitivity, one can check the triplosensitivity score determined by the ClinGen consortium or look for the pTriplo score available via the UCSC Genome Browser (Collins et al., 2022). We consider a curation from a reputable independent source, such as the ClinGen consortium, the highest level of evidence for triplosensitivity.

Allele Selectivity and Dosage Sensitivity

As noted, understanding the impact of dosage sensitivity helps inform which ASO strategy to pursue, including whether an allele-selective design must be considered. In diseases where complete loss of function is tolerated, targeting both alleles may be acceptable. However, if changes in gene dosage are known to cause disease, allele-selective ASOs, designed to target a SNP or the pathogenic variant itself, should be considered. While this approach can significantly constrain ASO design, it helps preserve wild-type function by selectively targeting only the disease allele, thereby theoretically maintaining at least 50% of normal protein expression. Depending on the ASO strategy, there are different considerations for allele selectivity, and are discussed in their respective sections where relevant.

Penetrance and variable expressivity

When assessing a variant's eligibility for therapeutic intervention, assessors should also take the potential for reduced penetrance and/or variable expressivity of the variant into account. To understand penetrance and variable expressivity, an assessor would need to better understand the clinical presentation of the case, which is outside the scope of this guideline. We, however, want to mention these concepts here for completeness. Our framework to assess genes and diseases addresses penetrance and variable expressivity in more detail.

In general, penetrance in genetics refers to the proportion of individuals carrying a given variant that also develop the associated phenotype. That means that carrying a variant with reduced penetrance does not necessarily mean the carrier will develop the phenotype. This should be taken into account in pre-symptomatic individuals for whom a therapy development is being discussed.

On the other hand, expressivity refers to the degree to which a phenotype is present in a given individual with a defined genotype (variant). That, for example, means that not all carriers of the variant will develop the full phenotype of the disease which will also inform the urgency of a therapeutic development.

Step 3 - Evaluation of splicing effects

In this step, assessors will evaluate the effect on splicing of each variant. As explained in the Background, correction of aberrant splicing is an elegant way to restore the reading frame and the physiological splicing pattern (Fig. 1). Thus, the first aim should always be to assess whether a splice-switching ASO strategy is applicable. Whenever this is not a possibility, the flowchart in Step 4 will aid in choosing the most suitable section for variant assessment. For the splicing evaluation and correction of aberrant splicing, different considerations have to be made given inheritance patterns and pathomechanism of the genetic variant (described separately at the end of Step 3). Table 3 provides an overview of the classification of variants for eligibility for ASO splice correction.

In this section, the following is covered:

1. Determining whether a variant affects splicing and what is considered sufficient evidence for mis-splicing
2. Considerations for eligibility of splice correction ASOs for intronic and exonic (including UTR) variants
3. Types of exonic variants that can cause aberrant splicing and alternate ASO strategies to consider in place of splice correction
4. Special considerations for canonical splice site variants
5. Important considerations for pathomechanism and inheritance pattern
6. How to search the literature for splice correction ASOs

Different types of variants can influence the splicing process and thus the decision on which ASO strategy to apply. While all variants should be assessed for their splice-altering potential (Abramowicz & Gos, 2018), some exceptions usually do not have to be evaluated in Step 3:

- For nonsense and frameshift variants that are associated with a LoF mechanism at the protein level, see Section A[7] (or Section C in cases of haploinsufficiency). Rare cases where nonsense and frameshift variants that affect splicing lead to GoF or DN effects can be assessed with the considerations outlined in Step 3.
- For whole exon duplications and deletions, please consider the variant effect and jump to Section B (knockdown) or Section C (upregulation from the wildtype), check Fig. 7 for directions.

The effects on splicing of each variant need to be confirmed with a functional assay, whereby only RNAseq, qPCR, or cDNA sequencing/analysis obtained from patient-derived cells can be considered sufficiently reliable. Please note that it is considered sufficient if functional data of the above-mentioned kind is available on the same variant from a different individual, i.e., a case report in the literature or information provided in ClinVar. Data gathered through artificial assays,

[7] One notable exception applies here. Nonsense and frameshift variants have a small theoretical probability of resulting in aberrant splicing (Haque et al., 2024). Depending on the splicing effect, this could be canonical exon skipping, in which case the variant is not eligible for exon skipping treatment. It could also be partial exon skipping/cryptic splicing in which the exon in which the variant is located can still be considered for canonical exon skipping, see Section A.

like exon-trapping and mini- and midi-genes, cannot be considered sufficient, as these assays do not recapitulate endogenous splicing, and results can be misleading (Lin et al., 2021). Explicitly, the prediction of splicing effects using *in silico* tools is not sufficient evidence and cannot be used for these assessments (Oh et al., 2024).
In case there is functional evidence against a splice-altering effect, i.e., confirmation that the variant does not affect splicing, please refer to Fig. 7 to guide you towards the next section applicable to your assessment. Additionally, if there is no functional evidence of splicing effects, please refer to Fig. 7. In cases where there is no evidence of splicing effects (whether for or against), this does not mean the variant is ineligible for a splice correcting ASO. A lack of functional evidence towards splicing means this variant cannot be assessed for splice correction eligibility at this time. Theoretically, all variants are suspicious of mis-splicing until demonstrated otherwise, and if evidence on splicing effects becomes available, the variant should be reassessed for eligibility towards splice correction ASOs.

If splicing effects are confirmed, the exact effects need to be evaluated. This can be a gain of an acceptor or donor splice site or the loss/weakening of an acceptor or donor splice site. Especially if canonical splicing is destroyed - due to canonical splice site variants, branchpoint variants, or variants destroying other splice-regulatory elements - the variant is most likely not eligible for a splice correction ASO treatment. Canonical splicing is considered destroyed if no wildtype transcript/splicing at the canonical splice sites can be identified in the functional analysis. If some wildtype transcript is still produced or protein function is detected, canonical splice sites are considered weakened. Variants that fully abolish canonical splicing are usually not amenable to a splice correction ASO and are classified as “not eligible” for this approach.[8] For other ways to target these variants, please see our section on canonical splice site variants.

We recommend studying available data carefully and assessing data provided in supplementary material, as one can often find relevant gels and blots in the supplement that may only be hinted at in the main manuscript. Further, look out for evidence illustrated by gels or qPCR results etc. instead of solely relying on how the authors describe or discuss the data. From our experience, some manuscripts claim there is no wildtype splicing left whereas faint wildtype bands can be identified in a gel, which might be sufficient to consider this a suitable case. Please consider that a variant's effect on splicing may not have been assessed. The absence of evidence might indicate that splicing effects were overlooked.

We generally distinguish intronic and exonic variants for the splicing assessments. Please also see Fig. 5 for a general overview of the classification of variants towards splice correction.

Intronic variants
As a rule of thumb, we consider intronic variants that are >-100 bp (upstream of acceptor splice site) or >+50 bp (downstream of donor splice site) away from the nearest canonical splice site as likely eligible, i.e., there should be no negative impact on the canonical splice sites and

[8] The assessment of whether the wildtype allele is still left can be difficult. For homozygous and hemizygous variants, this is straightforward, but in the case of heterozygous or compound heterozygous variants, this analysis is more difficult and information on the other allele is necessary to assess the effect of a variant.

branchpoint (Fig. 5)[9]. For variants closer to the canonical splice sites, enough functional evidence should be available on the effect on canonical splicing. Variants within 5 bp of the exon-intron/intron-exon boundary, often referred to as splice-site variants, are considered "not eligible" for splice correction.[10] There are some instances where splice-site variants lead to leaky exon skipping (Dawes et al., 2023; Feng et al., 2022). If functional data shows that the variant is causing leaky exon skipping, meaning that the exon is only skipped in some RNA transcripts, and the canonical splice site is weakened but not destroyed, please refer to the section titled "Considerations for Exon Inclusion" (Step 3) for further information. Note that the variant should also not disturb the branchpoint. The location of the branchpoint can vary and is usually found 18-40 bp upstream from the acceptor splice site, but can also be found anywhere 10-100 bp upstream (i.e., -10 to -100 bp region) (Xie, Wang, & Lin, 2023).

Intronic variants with functional evidence of causing aberrant splicing can now be analyzed using the above-mentioned criteria and classified using Table 3.

Exonic variants

For exonic variants, further considerations apply. As for the intronic variants, the exonic variant should be at least 5 bp outside of the canonical splice sites (upstream and downstream - hard cutoff). In some instances, a variant within 5bp of the canonical splice site may lead to leaky exon skipping, where the canonical splice signal is weakened but not destroyed. If functional evidence shows that this is the case, refer to the section titled "Considerations for Exon Inclusion" (Step 3). In cases where the variant is destroying the canonical splice site, leading to full exon skipping, these should be treated the same as a complete exonic deletion. Skipping an adjacent exon to restore the reading frame may be possible. For these cases, refer to Section A (exon skipping). We further define a second cut-off at 15 bp (region 6 to 15 bp upstream or downstream of the canonical splice sites) as a soft cutoff where a splice correction can be considered but is challenging. This cutoff is based on the idea that ASOs can bind to this region without destroying the canonical splice site, yet there is still a possibility of weakening canonical splicing.

[9] Please note again that a deep intronic variant in one transcript could be an exonic variant or canonical splice site variant in an alternative transcript. We recommend assessors to check this before continuing the assessment for a deep intronic variant.

[10] Avoiding +/-5 bp around the exon-intron/intron-exon boundaries is recommended because these variants are most likely destroying canonical splicing (or the splice site itself) which cannot be corrected. Additionally, it will be challenging to place an ASO within this region without negatively impacting canonical splicing. In a few cases, there are possibilities to target these variants. We discuss some of these considerations in "Considerations for Exon Inclusion".

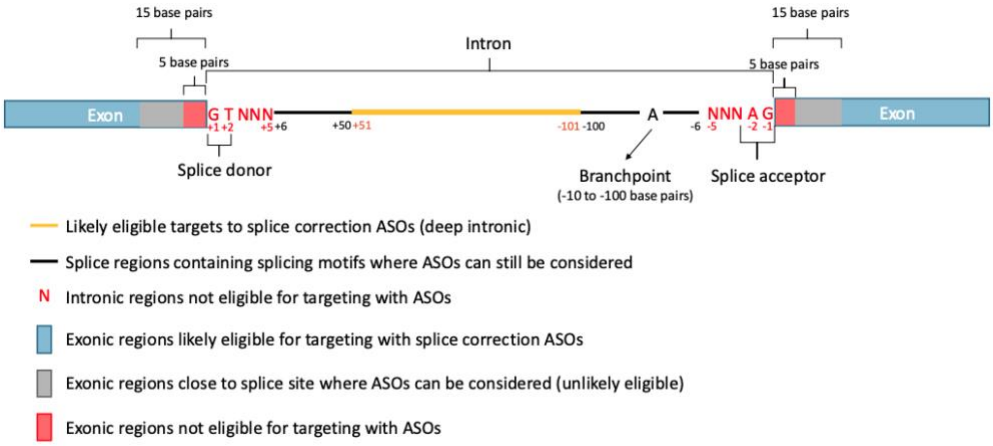


Figure 5: Eligible targets for splice correcting ASOs

The design of splice correcting ASOs takes into consideration key splicing motifs. The most amenable variants for splice correction are deep intronic variants, highlighted yellow in this figure. As the canonical splice sites are approached, one must consider the effects of the ASO blocking important splice site motifs (i.e., the branchpoint). These regions are highlighted in black (intronic) or gray (exonic) in this figure. Anything within 5 bp is considered not eligible for targeting with an ASO (highlighted red in this figure, both intronic and exonic), though exceptions apply.

For exonic variants, the specific type of variant is important for further analysis.

1. Nonsense and frameshift variants that cause aberrant splicing can be further distinguished:
    a) If the splice aberration itself leads to a LoF effect on the protein, the variant cannot be analyzed using this part of the guidelines. Correction of splicing would still lead to a LoF on the protein level. Thus, these variants fall under Section A for exon skipping analysis (or Section C in the case of haploinsufficiency).
    b) If the splice aberration leads to a toxic GoF effect on protein level, the variant can be considered for splice correction. Correction of the splicing effect will then lead to an early truncation and most likely a LoF effect on protein level. For certain cases, this is a useful ASO approach. It is also possible to consider these variants for downregulation (Section B).

    In both situations, our considerations for haploinsufficiency should be taken into account.
2. Synonymous variants that cause aberrant splicing do not influence the amino acid sequence and can be assessed using Table 3.
3. For missense variants and small in-frame indels (<50 bp) (Mahmoud et al., 2019), the considerations are more complex as both the effect on splicing and also the effect of altering the protein coding sequence need to be taken into account. It needs to be established that the variant effect at the protein level solely arises from the aberrant splicing and not that a missense variant or small indel itself causes a pathogenic effect. It is possible that a missense variant causing aberrant splicing leading to a LoF at the protein level, could independently cause a GoF or DN effect as a missense variant.

a) If the effect of a missense variant or in-frame indel on the protein sequence is not known (independent of the splice-altering effect), the variant cannot be assessed until further evidence is available.[11]
b) If the change in amino acid sequence is known to be pathogenic, for example, if a different pathogenic nucleotide change causes the same amino acid change without the splicing effect, different sub-criteria apply:
    i) Aberrant splicing and missense variant cause GoF/DN effect → can be considered for downregulation (Section B) or for exon skipping (Section A).
    ii) Aberrant splicing and missense variant cause LoF effect, with the amino acid change causing a complete loss of function on the protein level → can be considered for exon skipping (see Section A) or wildtype upregulation if a wildtype allele is available (see Section C).
    iii) Aberrant splicing causes GoF effect and missense variant would lead to a LoF → can be considered for exon skipping (Section A) or knockdown (Section B) and also for splice correction if the LoF phenotype is milder or loss of one allele is tolerated.
    iv) Aberrant splicing causes LoF effect and missense variant causes GoF/DN effect → can be considered for exon skipping (Section A) or for splice correction in case of a GoF/DN effect that leads to a less severe phenotype compared to the LoF phenotype.
    v) Aberrant splicing leads to GoF or LoF effect and the variant causes partial loss of function/reduced protein function → can be considered for splice correction and would be classified as “unlikely eligible” for splice correction. Here, the underlying thought is that having a partially functional protein is better than having no protein function or toxic protein function due to aberrant splicing. The variant can also be considered for exon skipping (Section A) or for downregulation (Section B) in case of GoF effects from aberrant splicing.
c) If the amino acid change is known to be benign on protein level, due to population data or functional studies, the variant follows the same considerations as a synonymous variant and can be analyzed using Table 3.

[11] One can consider exon skipping for the variant independent if the variant is LoF, GoF, or DN based on the amino acid change if it is possible to show that the resulting protein has some remaining typical function. Please see Section A.

UTR Variants

UTR Variants should be assessed in a manner similar to exonic coding variants. If there is functional evidence that a variant in the 5' or 3' UTR leads to aberrant splicing, they could be considered for splice correcting ASOs. For example, a 3' UTR variant in *KLHL40* resulted in the creation of a cryptic donor site and was assumed to cause nonsense mediated decay (Dofash et al., 2022). This variant could be treated as an exonic gain-of-splice variant and could be targeted with a splice-correcting ASO.

Important considerations for different inheritance patterns and pathomechanisms

For AR disorders, Step 3 applies without restrictions. If the variants are homozygous or compound heterozygous, the Step 3 evaluation has to be done once or twice for both variants respectively. If the disorder is AD associated with a LoF, the Step 3 guidelines also apply. Since the purpose of these ASOs is to restore wildtype splicing, rather than alter splicing, whether the ASO binds the wildtype or pathogenic allele does not matter (unlike with canonical exon skipping ASOs and gapmer ASOs). Similarly, variants that lead to a GoF or DN effect through altering splicing can be assessed with these guidelines (please consider the effect of the variant on protein level once splicing is corrected). For GoF or DN variants, however, one might consider a knockdown approach for a more efficacious effect (see Section B).

ASO Check

In addition to the recommended assessment strategies, assessors should review the literature for splice correcting ASO strategies. This review can be performed either as the final step to validate the assessment strategy or earlier in the assessment process (see Step 0). Specifically, for splice correcting ASOs, it is crucial to evaluate whether a splice correcting approach has been implemented and validated for the specific variant at the RNA level. Please also consider that the rescue at the RNA level needs to be sufficient to produce enough protein to rescue the phenotype. Ideally, a publication should provide this evidence.

Splicing mechanisms differ by variant, making it crucial to ensure that any existing ASOs found in the literature are applicable to the specific variant of interest. Additionally, if the literature indicates that an ASO is ineffective in correcting splicing for the variant of interest, this does not necessarily render the variant ineligible for ASO development. We recommend that a variant should only be considered "not eligible" for splice-correcting ASOs if there is evidence from two independent investigations at the protein or functional level, or from one investigation providing a convincing explanation as to why an ASO cannot be developed.

In cases of conflicting evidence, consider the quality of the research, the types of experiments conducted, the nature of the results shared, and the publication date. The evaluation of literature on existing ASOs should be carried out at the assessor's discretion, with a critical and discerning approach.

To help with identifying available ASOs for splice correction, we recommend a search term in Pubmed like this (searching for an ASO targeting a specific variant), text in bold would need to be adjusted for the gene and variant being examined:

ABCA4 *AND ((ASO) OR (AON) OR (antisense oligonucleotide)) AND ((**Gln876Ter**) OR (**c.2626C>T**) OR (**E876X**) OR (**Q876X**) OR (**Gln876***) OR (**E876***) OR (**Q876***))*

Table 3: Classification of variants for their eligibility towards splice correction.

Classification	Criteria
Eligible	ASO has already been developed and shown to work with available functional evidence at the protein level (pre-clinical data is sufficient)

Likely eligible	**Functional studies (RNAseq, qPCR)** validate alternate splicing
	
	AND (if intronic)
	
	{
	Intronic variant -101 bp or +51 bp outside of the canonical splice sites
	
	OR
	
	No weakened branch point/canonical splice site as determined by functional studies
	}
	
	AND (if exonic, including UTR)
	
	Donor/acceptor gained is not within 15 bp of a canonical splice site
	
	AND (if exonic coding)
	
	Functional evidence shows there is no pathogenic effect of the amino acid change(s) on the protein.

Unlikely eligible	**Functional studies (RNAseq, qPCR)** validate alternate splicing **BUT** { Canonical splice site and branchpoint is weakened (but still functional, i.e., either canonical transcript or protein function still detectable) **OR/AND (if intronic)** Intronic variant within -6 and -100 bp or +6 and +50 bp of the canonical splice sites. **OR/AND (if exonic)** Donor/acceptor gained within 6-15bp of the canonical splice site.

	OR/AND (if exonic) There is evidence of residual protein function (i.e., if splicing is corrected, the protein coding change still produces a partially functional protein). }
Not eligible	Canonical splice site and branchpoints are destroyed - functional evidence shows exon skipping in 100% of transcripts on the mutated allele **OR** Different nucleotide change leading to the same predicted amino acid residue change – but no alternate splicing – is pathogenic **OR** Variant within 5 bp of the canonical splice site **OR** Evidence that ASO cannot be developed, shown by two independent investigations on the protein/functional level **or** one investigation with a convincing explanation of why ASO cannot be developed.

Considerations for Exon Inclusion

Introduction to Exon Inclusion

As described earlier in Step 3, canonical splice site variants and variants disrupting splice regulatory elements (e.g., branchpoints, splice enhancers/silencer) are excluded from further assessment for splice correction, and are classified as not eligible. However, in some instances, weakened splice sites leading to exon skipping can be strengthened through blocking a splice silencer signal, meaning silencing the silencer will lead to increased exon inclusion. Exon

inclusion is the mechanism behind the most widely used ASO to date, nusinersen, that enhances the inclusion of exon 7 into the *SMN2* transcript (NM_017411.4) used to treat patients with spinal muscular atrophy (Hua et al., 2008). This is currently also the only drug in clinical application using this strategy, highlighting the difficulty in achieving sufficient exon inclusion. Further examples of successful pre-clinical exon inclusion have been published for *SLC26A4, PAH*, and *GAA* (Feng et al., 2022; Martínez-Pizarro et al., 2024; van der Wal et al., 2017).

In this section of Step 3, assessors will be given considerations on how to assess for exon inclusion. This approach is applicable for alternative exons that are rarely spliced in or exons that have a lower inclusion frequency due to pathogenic variants weakening the splice sites. The latter case is especially interesting as inclusion of these exons would lead to restoration of the canonical transcripts. However, this is where most often the challenge lies. Exons that are skipped due to pathogenic variants are incredibly difficult to include back into the transcript using ASOs. Therefore, it is not recommended to routinely perform exon inclusion assessments unless there is strong functional evidence supporting their feasibility, the investigator has sufficient technical expertise with this approach, and no other ASO strategy is applicable.

For completeness, we discuss how these assessments are generally performed. Unlike other ASO strategies discussed in these guidelines (splice correction, exon skipping, and knockdown) we do not assess eligibility for these variants here. We do however consider a variant eligible for exon inclusion in cases where an exon inclusion strategy has been published with clear functional data that goes beyond mini- and midi-gene studies, similar to considerations for wildtype upregulation (Section C).

Availability of Functional Data

To assess a variant for potential exon inclusion, functional data on the missplicing caused by the variant is necessary. There needs to be a clear understanding of the effect of the variant and the different transcripts produced. Preferably, this data is obtained from patient-derived cells. Please note that a mini-/midi-gene assay alone is not sufficient. Furthermore, the level of wildtype splicing needs to be established. Exon inclusion should only be considered for variants that weaken canonical splicing, either by weakening the canonical splice site or branchpoint directly or by disrupting a splice enhancer. Generally, there should still be substantial wildtype splicing present in the mutant transcripts. From the published literature where exon inclusion was shown preclinically, usually at least 20% (and up to 70%) of wildtype transcript was still present in the patients (Feng et al., 2022; Hua et al., 2008; Martínez-Pizarro et al., 2024; van der Wal et al., 2017). We thus recommend to ensure that sufficient levels of wildtype transcript are still produced.

Fully destroyed canonical splice sites or instances where canonical splicing is fully abolished cannot be rescued with exon inclusion strategies. Our explanations described earlier in this section (Step 3) are still true for these considerations here. These variants are considered “not eligible” for splice correction.

Exon Inclusion ASO Target Sites

When the exon skipping is caused by disruption of an exonic splice enhancer, exon inclusion can only be promoted by blocking an exonic or intronic splice silencer working in tandem. Similar considerations apply when exon skipping is caused by a canonical splice site variant. Such counteracting motifs are not always present.

Most often hnRNP regulatory motifs are being targeted, ideally in the intron. These motifs can be identified through searching for the sequences in the intronic region close to the respective splice site, usually an ASO walk is then performed to identify an ASO sequence that best blocks the silencer. Here, it should be ensured that the motif is not too close to the canonical splice site so that blocking the silencer with an ASO is not negatively impacting splicing on the canonical splice site. It should also be made sure that the splice silencer is indeed affecting the splice site in question and not regulating a different canonical or cryptic splice site. These analyses and studies are beyond the scope of these guidelines.

To summarize, for exon inclusion strategies, splicing data must be available from a patient-derived sample (endogenous splicing) that clearly indicates the levels of wildtype and missplicing. Should a significant amount of wildtype transcript still be present, we recommend reaching out to an expert group to inquire for assessment for exon inclusion strategies, including where to target.

Canonical Splice Site Variants and Splice Region Variants

As discussed above, canonical splice site variants and variants in close proximity to the canonical splice sites (with 5 bp) are usually not good candidates for an ASO strategy. The reason being that most often the canonical splicing is disturbed making it hard to correct these effects. Given the high demand for ASO therapies involving these pathogenic variants and the fact that exceptions do apply, we here discuss scenarios in which canonical splice site variants can be considered for an ASO approach. We recommend discussing such an approach with an expert before starting the development.

Pathogenic variants at or around the canonical splice sites can lead to a multitude of effects including full exon skipping, leaky exon skipping, intron retention, and cryptic splicing. It is also possible that a pathogenic canonical splice site variant leads to double- or even multi-exon skipping. An overview of potential effects is given in Fig. 6 and the matching ASO strategy is provided in Table 4. Please note that Fig. 6 is not exhaustive and combinations of effects can occur.

Before an ASO strategy can be chosen, functional data of the consequences of the variant need to be obtained that also include the level of wildtype transcript that is still present. Without such data, no proper strategy can be determined. In such cases, the variant remains “unable to assess”. Importantly, it can also be that a canonical splice site variant does not lead to splice disruption or minimal disruption not being disease-causing. Assessors should ensure that the variant is indeed the cause of disease before targeting such a variant.

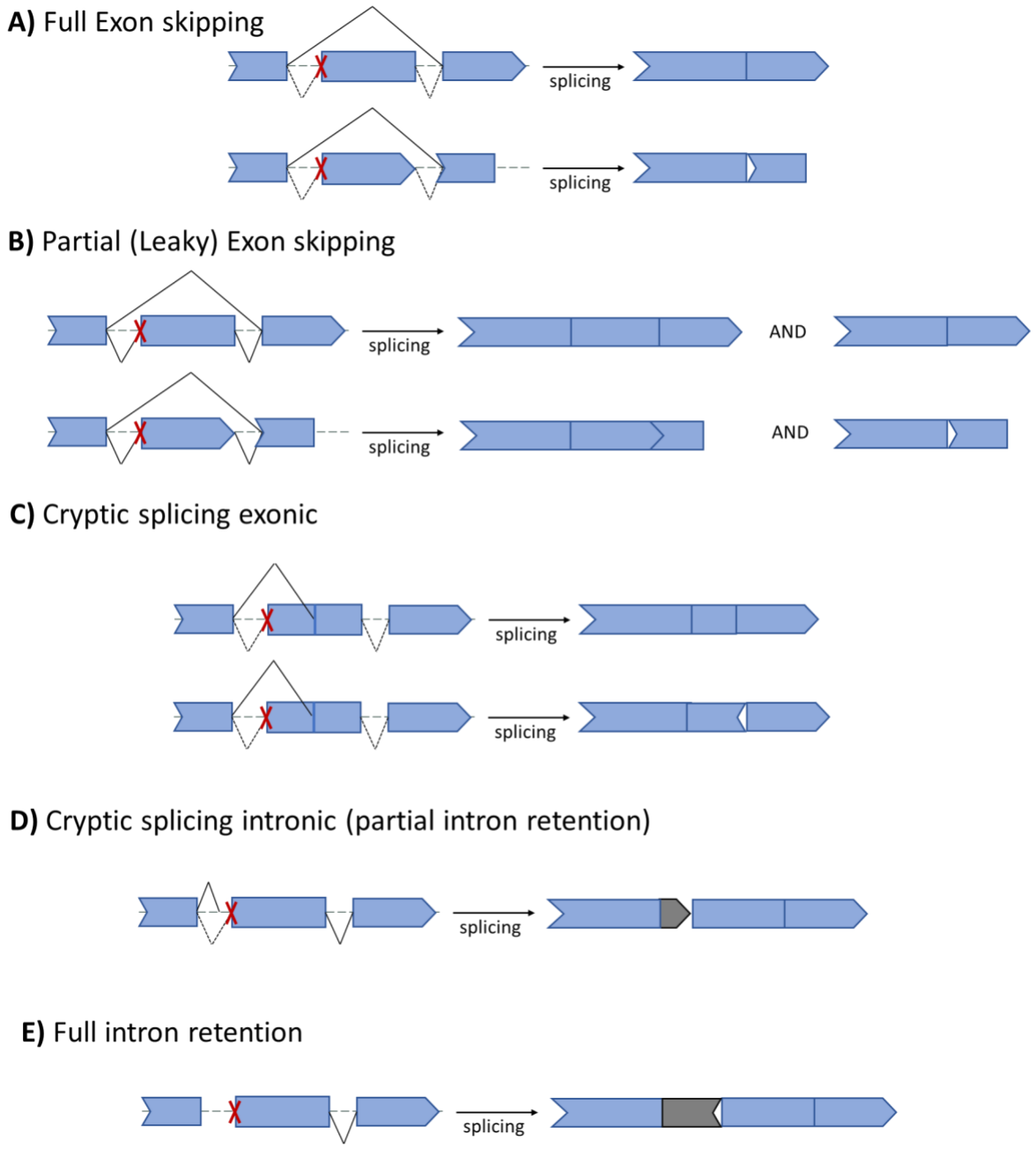


Figure 6: Possible aberrant splicing effects of canonical splice site variants

A) Full exon skipping can lead to different effects depending on whether the exon is in-frame our out-of-frame. **B**) In the case of leaky exon skipping, canonical mRNA transcript is still produced. **C**) In the case of cryptic exon splicing due to a canonical splice site variant, the cryptic splice site can be in-frame or out-of-frame with the canonical splice site leading to different effects. **D**) Partial intron retention can occur when intronic cryptic splice sites are used instead of the canonical splice site that is disrupted. In some cases (**E**), full intron is retained. In cases C-E, the effects can also be leaky with some canonical splicing still being present.

Table 4: Overview of consequences of canonical splice site variants and potential therapeutic strategies

Effect on splicing	In-frame exon	Out-of-frame exon
Full exon skipping	Not eligible for splice correction or exon skipping approaches, consider upregulation from the wildtype in case of LoF with haploinsufficiency or knockdown if the effect is DN or GoF.	Can be treated as a single-exon deletion and adjacent exons can potentially be skipped. Further consider upregulation from the wildtype in case of LoF with haploinsufficiency or knockdown if the effect is DN or GoF.
Partial (leaky) exon skipping	If sufficient WT splicing is left one can consider exon inclusion IF the effect can be counteracted by blocking a strong splice silencer. Further consider upregulation from the wildtype in case of LoF with haploinsufficiency or knockdown if the effect is DN or GoF.	
Cryptic splicing exonic	Not eligible for splice correction or exon skipping as the loss of an in-frame part of the exon leads to the phenotype. Consider upregulation from the wildtype in case of LoF with haploinsufficiency or knockdown if the effect is DN or GoF. If cryptic splicing is leaky, see leaky exon skipping.	Can be considered for full exon skipping (Section A) in case where the exon itself is in-frame and fulfills the criteria for exon skipping. Consider upregulation from the wildtype in case of LoF with haploinsufficiency or knockdown if the effect is DN or GoF. If cryptic splicing is leaky, see leaky exon skipping.
Cryptic splicing intronic/partial intron retention	Usually not eligible for splice correction if the canonical splice site is fully destroyed. If splicing is leaky, consider blocking the cryptic splice site with an ASO or try to enforce full exon skipping. Consider upregulation from the wildtype in case of LoF with haploinsufficiency or knockdown if the effect is DN or GoF.	Usually not eligible for splice correction if the canonical splice site is ffully destroyed. If splicing is leaky, consider blocking the cryptic splice site with an ASO. Consider upregulation from the wildtype in case of LoF with haploinsufficiency or knockdown if the effect is DN or GoF.
Full intron retention	Usually not eligible for splice correction and exon skipping. Consider upregulation from the wildtype in case of LoF with haploinsufficiency or knockdown if the effect is DN or GoF.	

Step 4 - Identification of Relevant Guideline

In this step, assessors will be guided towards the sections (sub-guideline) applicable to the variant under assessment. Once the inheritance pattern, pathomechanism, and splicing effects are evaluated, a decision on the most useful guideline(s) can be made. If a variant does not have an effect on splicing, or an effect on splicing that cannot be assessed using Step 3, the assessor can turn to the below flow diagram and chart for assistance.

The table within the flow chart lists all applicable sections and with the knowledge of the inheritance pattern and pathomechanism, the assessor can now identify the applicable section(s) to be read to make the assessment.

Please note again that each section is a stand-alone sub-guideline and thus one can jump to the necessary section without having to read the full text.

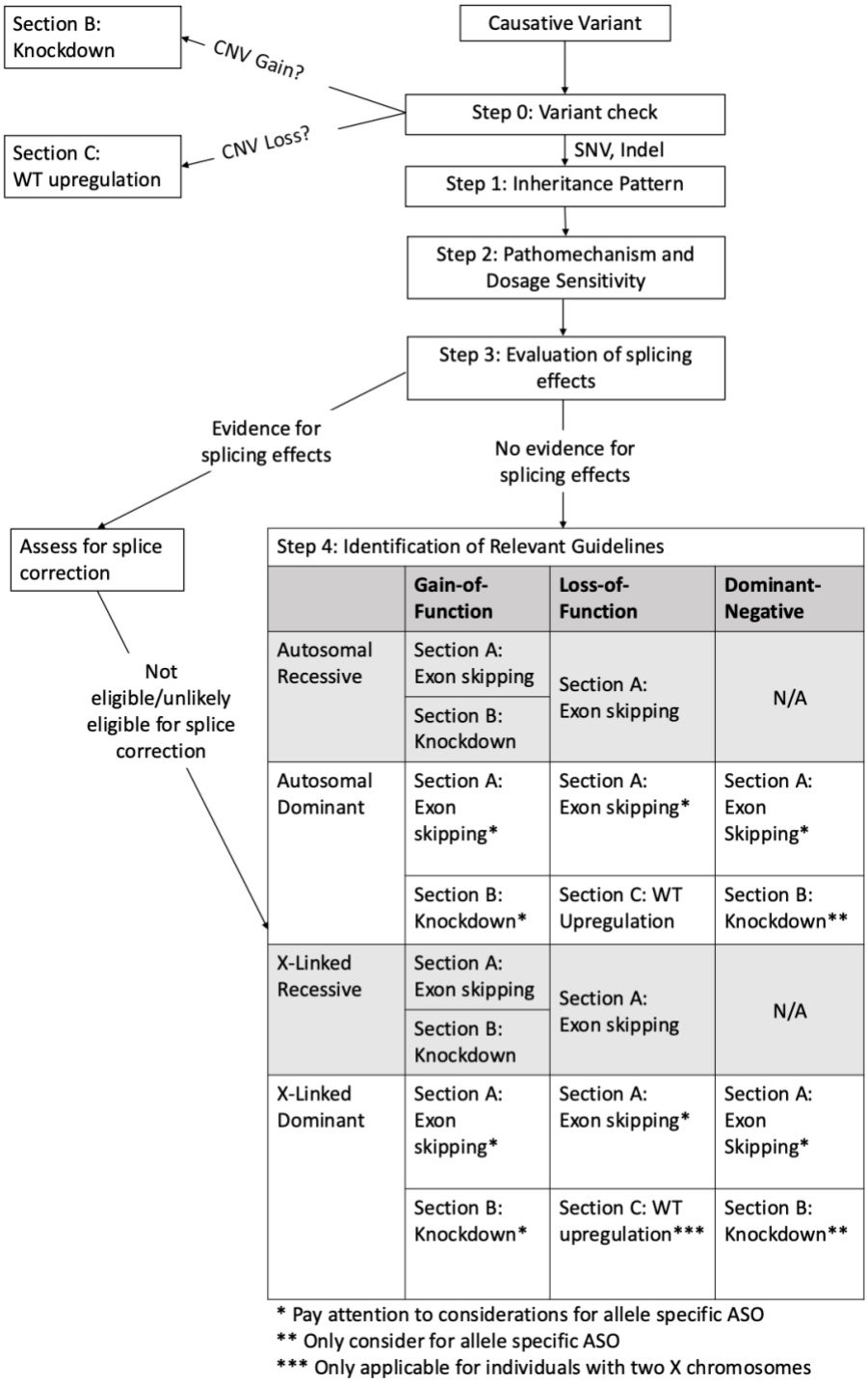


Figure 7: Flowchart for the identification of relevant guidelines.
Steps 0, 1, and 2 require the assessment of the variant description, inheritance pattern, and pathomechanisms. The evaluation of splicing effects is considered in Step 3. Depending on the information gathered in Steps 0-3, one can consider the guidelines discussed in Sections A, B or C.

Section A - Considerations for Canonical Exon Skipping

In this section, assessors will be guided on how to assess a variant for eligibility towards canonical exon skipping ASOs. As described in the Background, exon skipping ASOs can be used to skip exons containing pathogenic variants (Fig. 1). Variants generally to be considered for exon skipping are nonsense and frameshift variants causing a LoF in AR, AD, X-linked and Y-linked disorders (additional considerations apply in dominantly inherited diseases, see "Important considerations for different inheritance patterns and pathomechanisms" at the end of this section). GoF and DN negative variants can also be amenable for an exon skipping approach under certain circumstances as well as whole exon deletions (see "Important considerations for different inheritance patterns and pathomechanisms" in this section). While splicing assessments are based on the variant, evaluating the potential for exon skipping are based on the exon, thus an exon skipping approach is applicable to all variants within that exon. For an exon to be skippable, different criteria need to be met (Fig. 8).

This section covers the following topics:

1. Assessing exon position and frame
2. Assessing exon size
3. Strategies for searching and considering naturally occurring exon skipping and in-frame deletions
4. Assessing the role of functional domains
5. Important considerations for different inheritance patterns and pathomechanisms
6. Strategies for searching the literature for exon skipping ASOs



Figure 8: Overview of exon skipping assessment using a hypothetical transcript.
The first and last coding exons (exons 2 and 10) cannot be skipped, in addition to out-of-frame exons (exons 3,4,7,9, shown by the shape of the exons). In-frame exons encoding for more than 10% of the coding region (exon 5) and exons coding for functional domains (exon 8) are unlikely eligible for exon skipping. Likely eligible for exon skipping are small, in-frame exons that do not code for a domain (e.g., exon 6). White: exon not considered for canonical exon skipping, e.g., exons in the UTRs, grey: exons not eligible for skipping, blue: exons unlikely eligible for exon skipping, green: exon likely eligible for exon skipping, orange: functional domain.

Assessing Exon Position and Frame

The first and last coding exons are usually not eligible for exon skipping as this would lead to the loss of the start or stop codon. Note, due to untranslated regions (UTRs) the first and last exons of a transcript are not necessarily coding exons and, thus, assessors should be aware of which exons contain the canonical start and stop codons (Aspden, Wallace, & Whiffin, 2023) (Fig. 8). Additionally, genes containing exactly one coding exon are not eligible and are disqualified from

further analysis.[12] Next, assessors should evaluate the length and position/frame of the exon. Out-of-frame exons (containing a number of base pairs not divisible by 3) should be classified as "not eligible" in the case of LoF variants where restoration of the reading frame is the aim. Assessors can use tools such as the UCSC Genome Browser, ExonViz (van den Berg, Lauffer, & Laros, 2024), or Ensembl to determine whether the exon is in-frame or out-of-frame and whether the exon is coding or non-coding (Fig. 9).

For in-frame exons, it is possible that the first and last codon is partially encoded by the adjacent exons (exons with phases 1-1 and 2-2). In this case, the formation of a new codon (including stop codons) is possible at the new exon-exon boundary. Assessors should evaluate whether joining the adjacent exons would form an in-frame stop codon, coded by either a TAA, TAG, or TGA (Fig. 10). If this is the case, the exon is not eligible for exon skipping (exceptions apply, please see below). If a novel codon is created that results in a different amino acid at the same residue, we recommend classifying this as unlikely eligible. Assessors can utilize the UCSC Genome Browser or Ensembl to check (instructions on how to perform this check are provided in example video 11, the NM_000170.3(GLDC):c.538C>T [MIM 238300] example).

[12] One notable exception is if these exons contain GoF variants. By skipping the only coding exon, one would remove the GoF variant. This can be seen as a knockdown approach and other criteria for knockdown approaches need to be met for this strategy to be useful.

A)



B)

Show All entries Show/hide columns

No.	Exon / Intron	Start	End	Start Phase	End Phase	Length
	5' upstream sequence					
1	ENSE00002033907	166,128,020	166,127,771	-	-	250
	Intron 1-2	166,127,770	166,127,011			760
2	ENSE00003792641	166,127,010	166,126,924	-	-	87
	Intron 2-3	166,126,923	166,077,802			49,122
3	ENSE00002040027	166,077,801	166,077,710	-	-	92
	Intron 3-4	166,077,709	166,073,671			4,039
4	ENSE00003761695	166,073,670	166,073,358	-	0	313
	Intron 4-5	166,073,357	166,058,689			14,669
5	ENSE00002470479	166,058,688	166,058,570	0	2	119
	Intron 5-6	166,058,569	166,056,501			2,069
6	ENSE00001785173	166,056,500	166,056,411	2	2	90
	Intron 6-7	166,056,410	166,054,767			1,644
7	ENSE00001286304	166,054,766	166,054,638	2	2	129
	Intron 7-8	166,054,637	166,052,944			1,694
8	ENSE00002441017	166,052,943	166,052,852	2	1	92
	Intron 8-9	166,052,851	166,051,989			863
9	ENSE00001128613	166,051,988	166,051,719	1	1	270

C)

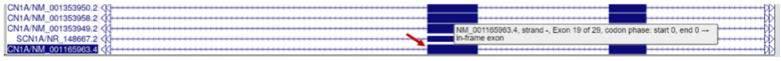


Figure 9: Determining exon frames.
A) Exon frames in ExonViz can be identified by their shape. Rectangular shapes are in-frame with a phase at 0-0 (exon starts and ends with the codon). Exons with an arrow on one end and a notch on the other end are also considered in-frame, whereby it does not matter if the arrow is upstream or downstream. If there are two arrows or two notches, the exons are out-of-frame. An arrow indicates a 1 nucleotide overhang while a notch illustrates a 2 nucleotide overhang. Small, in-frame exons are circled in red (exon 6, 7, 17 and 25). For example, exon 6 starts with an arrow, meaning the exon starts with 1 nucleotide from a codon starting in the previous exon (i.e., this is the third nucleotide in the codon, with the first 2 nucleotides being found in exon 5). Exon 6 ends with a notch, indicating that these are the first two nucleotides of a codon (2-2 phase). In total, both ends together cover 3 nucleotides. **B)** Identification with Ensembl can be done by checking the phases of an exon. Exons with phases 0-0, 1-1, 2-2 are in-frame. Additionally, dividing the exon length by 3 can help determine the frame. **C)** UCSC Genome Browser shows phases to determine the frames using mouse over. The browser directly indicates the phase for you. Please ensure you are hovering over the correct transcript.

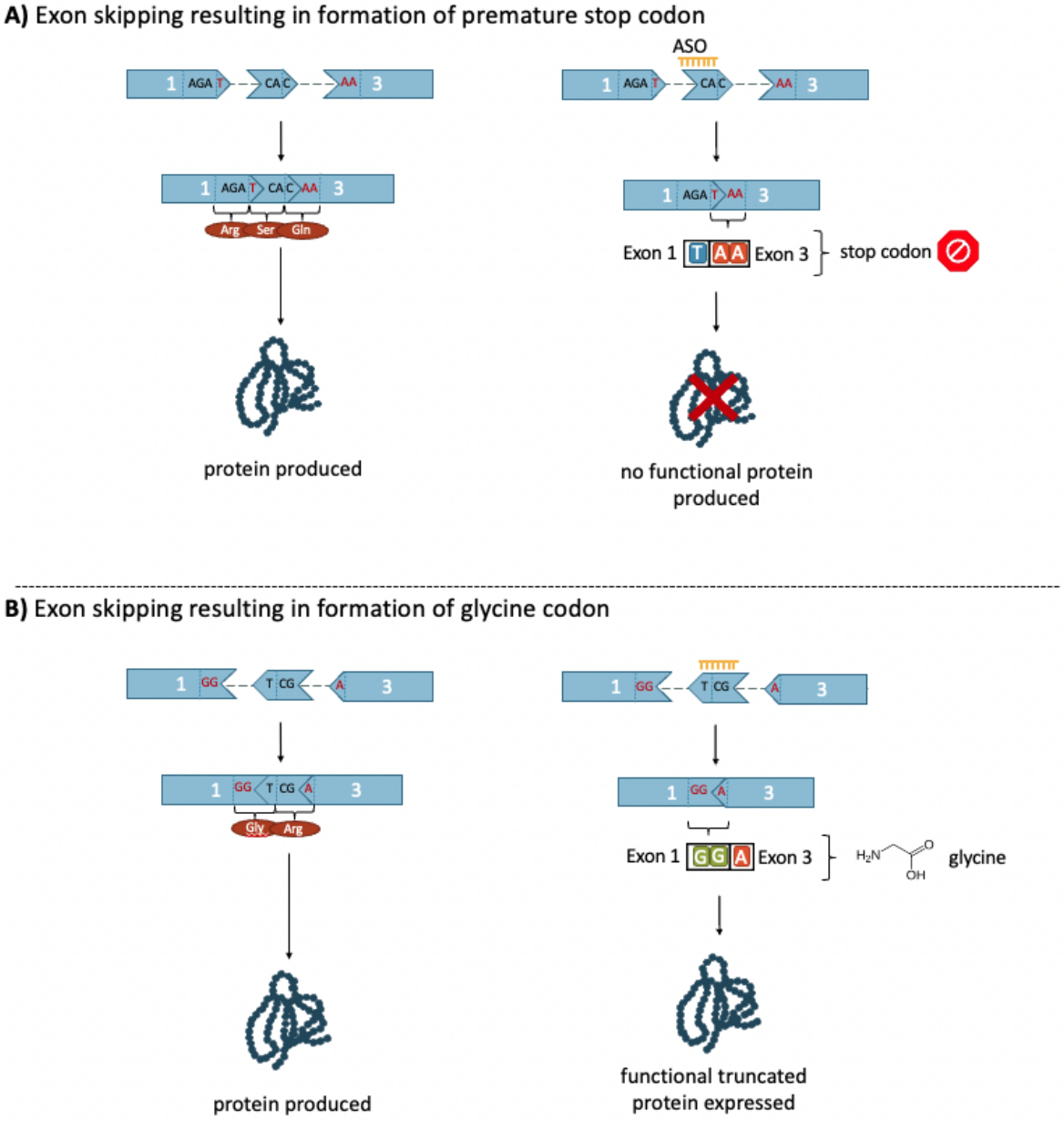


Figure 10: Formation of a new codon as a result of exon skipping.
When an in-frame exon with phase 1-1 or 2-2 is skipped, the adjacent exons will form a new codon on the new exon-exon boundary. **A)** shows the formation of a stop codon when an ASO is used to skip in-frame exon 2. The "T" nucleotide from exon 1, and "AA" nucleotides from exon 3 join together to form a stop codon. This leads to a premature termination resulting in an absent gene product or a truncated, non-functional product. **B)** shows the formation of a new codon with a functional product. The "GG" nucleotides at the end of exon 1 join the "A" nucleotide at the start of exon 3, upon skipping of exon 2. This codes for glycine.

The skipping of an exon that disrupts the reading frame (e.g., skipping of an out-of-frame exon, formation of a new termination codon when an exon is skipped, or skipping the first and last coding exon) tends to be not eligible for analysis. However, there are exceptions to these rules:

- It is theoretically possible to skip out-of-frame exons to generate a premature termination codon within the last or penultimate exon given that there is some left-over function of the protein (please note that this is theoretically possible - one would still have to assess the last exon for the importance of domains and whether this shortened transcript is predicted to undergo nonsense mediated decay)
- It is also possible to skip out-of-frame exons for GoF or DN variants to downregulate transcript levels, please see "Considerations for different inheritance patterns and pathomechanisms" in this section, as well as Section B
- It is theoretically possible to skip an in-frame exon which results in the formation of a stop codon if the resulting protein is stable and has some left-over function (unlikely eligible)
- It is possible to skip adjacent out-of-frame exons in case of the variant being a whole exon deletion of an out-of-frame exon which will ultimately result in restoration of the reading frame. Yet, the other criteria on the size of the skipped area (now deletion + skipped exon) and functional domains apply (see section titled "Notes on multi-exon Skipping")
- It is theoretically possible to skip consecutive out-of-frame exons so as not to disrupt the reading frame, though designing an ASO that skips two consecutive exons may prove to be a challenge
- It is theoretically possible to skip the first coding exon if a nearby in-frame start codon exists, and the first exon meets exon skipping criteria outlined in Table 5

Assessing Exon Size

Assessors should consider the size of the exon. As per ClinGen recommendations, an in-frame deletion in the size of 10% or more of the coding transcript is considered a strong criterion for loss of protein function (Walker et al., 2023). However, this is protein and exon-dependent (i.e., losing up to 30% of the dystrophin transcript can still result in a functional, truncated protein (Duan, 2016; Gao & McNally, 2015)). For this reason, skipping an exon that encodes for more than 10% of the protein is considered "unlikely eligible", and obtaining functional evidence is the expected next step.

The percentage of coding region can be calculated as follows:

*Exon size as coding region in % = (exon length in bp/3) / length of the protein in aa *100*

Please see the example for the NM_003793.4(CTSF):c.264del (p.Cys89fs) [MIM 603539] variant in example video 10. The protein is 484 aa in length. Exon 2 is in-frame 0-0 and spans 99 nucleotides. This information can be obtained from Ensembl, UCSC genome browser, and/or UniProt (see video).

Exon size in coding region % = (99/3) aa / 484 aa * 100 = 6.8 %
Alternatively, assessors can use the Genetic Therapy Generator Toolkit from the Dutch Center for RNA Therapeutics that automatically calculates the exon length for you. By submitting the

variant in question, the GTGT will indicate how much protein coding region is left following skipping of the exon.

There are instances where an exon coding for more than 10% of the protein should be classified as “not eligible”. If the exon encodes for more than 10% of the protein and codes for more than one non-repeat protein domain, this exon should be considered non-skippable and therefore not eligible for exon skipping ASOs. For more detail on assessing the role of functional domains, please see the section titled “Assessing the Role of Functional Domains”.

In addition to the relative size (coding percentage), it is also important to consider an exon’s absolute size in nucleotides, as even large exons may constitute only a small portion of a very large protein. Large exons are generally more challenging therapeutic targets for ASO-mediated skipping because they often contain multiple splicing regulatory elements and cryptic splice sites, in theory resulting in the need for multiple ASOs to achieve effective exon skipping. An example is an ASO targeting exon 9 of the *ATXN3* gene which ended up promoting a partial exon 9 skip via the activation of an alternative 5′ cryptic splice site (Evers et al., 2013). Instead, current successful exon skipping ASOs target moderately sized exons typically less than 300 nt, such as DMD exons 45, 51, and 53. A length less than 300 nt is believed to facilitate exon definition and recognition by the spliceosome, minimizing the likelihood of additional splicing regulatory elements and internal cryptic splice sites (Bolisetty & Beemon, 2012). Due to this, it is recommended that assessors be cautious when considering ASO-mediated skipping of exons of 300 nt or more.

To further refine exon eligibility toward skipping, assessors can also use the recently developed VulExMap webtool, which classifies exons as either resilient or vulnerable to aberrant splicing as a result of mutations (Holm et al., 2024). Vulnerable exons, associated with less enforced inclusion and sensitivity to splicing perturbations, may be more amenable to ASO-mediated skipping, whereas resilient exons are robustly included in transcripts and may be more difficult to skip.

Searching for Cases of (Natural) Occurrences of Exon Skipping or In-frame Deletions

Assessors should determine whether exon skipping has already been observed naturally. This information is necessary to further determine whether exon skipping is a suitable option for a given variant. There are different ways in which exon skipping can occur naturally and resources such as ClinVar, gnomAD, DECIPHER, ExonSkipDB, and PubMed aid with their identification:

A) Canonical splice site or splice region variants
   Assessors should search ClinVar, PubMed, gnomAD, or DECIPHER to collect data on variants that cause splice aberration leading to full exon skipping. This information should have been validated by sufficient functional data (e.g., RNAseq or qPCR). Note that variants affecting splicing motifs outside of the splice sites can also cause exon skipping and would also fall under this category.

If full exon skipping has been observed and validated and is disease-causing (pathogenic variant), this exon (and therefore variants within that exon) is not eligible for exon skipping therapy. Please note that *in silico* predictions are not to be used as substitutes for RNAseq and qPCR data, and assessors must pay careful attention to what tools were used in the assessment of splicing outcomes of canonical splice site variants.
If full-length exon skipping has been observed in individuals who do not show signs of the disease in question (individuals can of course have other diseases) the exon is eligible for exon skipping. If no evidence of pathogenic or benign exon skipping validated by qPCR or RNAseq exists, assessors should proceed with the analysis.

B) Full exon deletions or in-frame deletions within the exon
Assessors should search ClinVar, PubMed, gnomAD, or DECIPHER to collect data on in-frame deletions of full-exon deletions within the exon of interest. If in-frame deletions have been observed and validated to be disease-causing (pathogenic), this exon (and therefore variants within that exon) is not eligible. If this is the case, no further analysis is required. However, assessors should pay attention to whether the in-frame deletion creates a stop codon or new amino acid, as these exons could still be eligible for exon skipping via ASO (i.e., the pathogenicity is possibly a result of premature termination or the change of an amino acid leading to folding changes and not necessarily the deletion of amino acids itself). If this is the case, assessors should proceed with analysis.
If full exon deletions and in some instances larger in-frame deletions are identified in the general population and classified as benign, this exon can be considered “eligible” for an exon skipping approach.

The assessment for the occurrence of natural exon skipping can easily be summarized as follows:

Eligible - Loss of the exon, either due to splice aberration or genomic deletions, has been classified as benign

Not eligible - Loss of the exon or larger parts of the exon have been classified as pathogenic

Assessing the Role of Functional Domains

Assessors should also consider the functionality for which the exon codes. To begin, assessors should identify which domains are coded by the exon using tools such as the UCSC Genome Browser. Please cross-check domain annotations with UniProt or other sources to ensure the correct protein isoform is tagged. In this regard, we consider all types of domains as domains, including transmembrane and cytoplasmic domains as well as disordered regions. At this point, we do not have enough information and data to distill which domains can be removed without negative impact, yet, there are certain types of domains for which we can assume they are more important than others and of such high importance that skipping is not an option. Assessors can search for the role of the amino acids or domains present in an exon in the literature using PubMed, or utilize databases such as the Protein Database (PDB) or UniProt. Assessors should pay attention to the role of these specific domains (i.e., DNA binding domain, catalytic domains) or amino acids (i.e., specific amino acids known to play an important role in enzyme activity or

protein structure). Sometimes, a web-search with “[protein name] protein structure” will also yield the desired results.

Though it is difficult to define the importance of the domain or to predict the effects of exon skipping on protein function, assessors can consider the following exons as “not eligible” for exon skipping:

- The exon codes for important amino acid(s) with a known key functional role in the protein, or a functionally validated domain (i.e., involved in a catalytic domain, dimerization domain, inhibitory domain, etc.)
- The exon is a known mutational hotspot for pathogenic LoF missense variants (tools such as MetaDome, Essential3D, and Franklin can be used to help determine this). There is no strict definition of a mutational hotspot, but one can look at the distribution of the variants across the exons and specifically pay attention to the number of missense variants causing loss of protein function in a specific region/exon as this is highly indicative of an important functional domain. Further, if the ratio of LoF missense variants is higher than that of LoF truncating variants, this is a further indicator for an important functional domain (see also below “Special considerations for missense variants and small in-frame indels”). Web resources such as Franklin also have some calculations to indicate a mutational hotspot.
- The exon codes for the only functional domain in the protein (i.e., no possibility of residual function remaining)
- The exon codes for multiple functional domains and covers more than 10% of the coding region

An exon that codes for a functional domain that does **not** meet the above criteria means it can be considered as “unlikely eligible”, indicating the need for functional studies. It is of note, however, that emerging research from newly developed ASOs has revealed exceptions to the above criteria. The ASO therapy zebronkysen, which promotes exon 8 skipping in the *CLN3* (NM_001042432.2) gene to restore protein function as a potential treatment for Batten disease, has yielded promising preclinical and clinical results despite skipping an exon that encodes for a transmembrane domain (Cotman & Staropoli, 2012). This development reiterates the importance of reviewing literature on each exon of interest and the significance of functional studies following assessment. Nevertheless, it is still recommended that exons encoding for important functional domains not be considered high priority, and only functionally assessed if resource availability permits it.

The above considerations apply whether the exon encodes part of a domain or the entire domain.

Additionally, **protein tandem repeat domains** can also be considered as “unlikely eligible”, assuming the loss of a repeat (or part of a repeat domain) from the protein will still have a residual function (Duan, Goemans, Takeda, Mercuri, & Aartsma-Rus, 2021). Assessors can utilize the aforementioned strategies to assess tandem repeat domains. However, in cases where in-frame deletion of a repeat domain has been reported as pathogenic, or has been functionally proven to disrupt protein function, these exons would be considered “not eligible”. Most suitable are in-frame

exons that contain a single, full repeat of the tandem repeats and no additional domains. For exon skipping of tandem repeat domains, we recommend that the protein consists of at least 5 tandem repeats. To identify exons that fulfill these criteria, assessors can utilize the tool TReXome DB.

Generally, we encourage contacting experts on a given gene and protein to discuss the role of the different domains and whether they consider exon skipping a possibility. Further, when looking into functional domains please check the exact role of a domain and the effect of losing the domain. For example, skipping an inhibitory domain could lead to a gain of function effect on protein level.

If the exon codes for less than 10% of the protein, and no functional domain or important amino acids are coded for by the exon, and the exon fulfills other criteria listed in Section A, then the exon can be considered as “likely eligible”. Exons are classified using the criteria outlined in Table 5.

Special considerations for missense variants and small in-frame indels

If a missense variant or small, in-frame indel leads to a LoF on the protein level, additional caution is necessary. These types of variants often indicate that the exon has a specific function even though no domain might yet be annotated. This could be that the respective amino acids are crucial for folding or the function of a certain domain has not yet been established. In these cases, we recommend paying special attention to other reported pathogenic variants within the exon. Should there be more pathogenic missense variants and in-frame indels than truncating variants, we consider this exon “unlikely eligible” for exon skipping.

However, for missense variants within protein tandem repeat domains, we refer to the criteria outlined in the tandem repeat domain section above as this is an exception to this recommendation.

Notes on multi-exon skipping

Double- and multi-exon skipping have been suggested as approaches to bypass deleterious variants in out-of-frame exons when skipping adjacent exon(s) maintains the open reading frame. While this has been explored in preclinical studies, such as when targeting truncating variants in *DMD* exon 43 via 43-44 skipping (Aartsma-Rus et al., 2004), we currently do not recommend such an approach for clinical practice as it often requires the use of multiple ASOs.

However, restoration of the reading frame in case of deletion of one or more out-of-frame exons can be achieved via skipping of a second out-of-frame exon that is adjacent to or downstream from a disease-causing exon deletion. If considering such an approach, criteria on exon size of the skipped area (including deleted + skipped exons), functional domains, and presence of missense variants/small in-frame indels still apply. In addition to this criteria, assessors should also consider utilizing tools such as GTGT, ExonViz, and the UCSC Genome Browser to identify instances where double- and multi-exon skipping produces an in-frame transcript. Assessors can

also search for existing cases of double- or multi-exon deletions in the target region that are associated with mild to asymptomatic phenotypes as rationale for such an approach.

Important considerations for different inheritance patterns and pathomechanisms
The recommendations described above apply without restrictions to LoF variants in recessive disorders. For LoF variants in AD disorders, one might consider the development of allele-selective ASOs. This will ensure that exon skipping does not occur on the wildtype transcript and may result in greater amounts of functional gene product compared to non-allele selectiveASOs. Though specific ASO design is beyond the scope of these guidelines, this may prove challenging and limit the types of ASOs to be designed.

In the specific case of an AD disorder with an out-of-frame exon deletion, allele-selectivity is necessary otherwise exon skipping is not possible. Here, exon skipping of adjacent out-of-frame exons has the potential to restore the reading frame on the mutant allele, at the same time, exon skipping of that out-of-frame exon would destroy the reading frame of the wildtype allele.

Further, for AD disorders associated with LoF variants, an upregulation of protein production from the wildtype allele should also be considered as an alternative option to exon skipping (see Section C).

For GoF variants, additional considerations are possible. The guidelines for exon skipping can be applied as they are, with the exception that toxic GoF caused by the disruption of an inhibitory domain will not be rescued by skipping the exon containing the inhibitory domain (Mohassel et al., 2021). For GoF variants, an out-of-frame exon can be skipped, either the one containing the variant or any other one to downregulate transcript levels. In such cases, haploinsufficiency of the gene should be taken into account, which might require allele-selective approaches. Generally, for the downregulation of a transcript in the case of GoF and DN variants, Section B can be consulted. We do not recommend skipping out-of-frame exons for DN variants unless the WT allele remains intact (allele-selective skipping). For DN variants, it is important that the ratio of WT to mutant allele needs to be shifted towards more WT allele.

ASO Check
In addition to the recommended assessment strategies, assessors should review the literature or registries for exon skipping ASO strategies. This review can be performed either as the final step to validate your assessment strategy or earlier in the assessment process (see Step 0). Specifically, for exon skipping ASOs, it is crucial to evaluate whether an exon skipping approach has been implemented and validated at both the RNA and protein levels and shown to have the desired effect on the phenotype.

In cases of conflicting evidence of ASO developments and the feasibility of an ASO therapy, consider the quality of the research, the types of experiments conducted, the nature of the results shared, and the publication date. The evaluation of literature on existing ASOs should be carried out at the assessor's discretion, with a critical and discerning approach.

For exon skipping ASOs specifically, the approach will be applicable to all variants within a given exon. However, please pay attention to whether an existing ASO would bind to the variant site and could thus not be used for the individual under assessment if it is a different variant. Also check if the existing target site contains a SNP and whether the same SNP is present on the correct allele in the case under assessment. The exon (and therefore variant) is still considered "eligible" for canonical exon skipping according to the criteria outlined in Table 5 even if the ASO binds to a different variant site. This, however, means that a new ASO needs to be designed and developed for the variant under assessment.

To help with identifying available ASOs for exon skipping, we recommend a search term in Pubmed like this, text in bold would need to be adjusted for the gene and exon being examined: ***ABCA4*** *AND ((ASO) OR (AON) OR (antisense oligonucleotide)) AND (****Exon 17****)*

Table 5: Classification of variants for their eligibility towards exon skipping.

Classification	Criteria
Eligible	Evidence that exon skipping does not impair protein function (benign canonical splice site variant leading to exon skipping, benign single exon deletion, naturally occurring transcript does not contain exon, or previously tested ASO with evidence of functional protein product).
Likely eligible	Exon is in-frame **AND** Exon does **NOT** result in ≥10% loss of protein coding sequence if exon is skipped. **AND** Exon does not create a stop codon **OR** new codon when skipped **AND** Exon does not code for any functional domains **AND** None of the exclusion criteria outlined in the "Not eligible" section are met.

Unlikely eligible	Exon is in-frame
	
	AND/OR
	
	Exon does not create a stop codon when skipped **BUT** creates a new codon
	
	AND/OR
	
	{
	Exon results in a loss of ≥10% of coding transcript if exon is skipped **AND/OR** exon codes for a <u>single</u> functional domain
	}
	
	AND
	
	None of the exclusion criteria outlined in the "Not eligible" section are met

Not eligible[13]	Variant is in an out-of-frame exon **OR** Variant is in first or last coding exon **OR** Variant is in the ONLY coding exon **OR** Exon skipping results in a stop codon **OR** Exon skipping results in the loss of the ONLY functional domain in the protein **OR** Exon encodes for more than 10% of the protein AND multiple non-repeat domains **OR** Exon codes for functionally proven important domains or amino acids (catalytic site, dimerization domain, inhibitory domains, etc.) **OR** Exon is a known mutational hotspot for (missense) loss-of-function variants **OR** Functional evidence of exon skipping shown to be pathogenic **OR** Functional evidence of in-frame deletions shown to be pathogenic **OR** Evidence that an ASO cannot be developed, shown by two independent investigations at the protein/functional level or one investigation with a convincing explanation why an ASO cannot be developed.

[13] Exceptions apply here for GoF and DN variants. Please see full text.

Section B - Considerations for Transcript Knockdown

In this section, assessors will be guided on how to assess a variant for eligibility towards knockdown approaches. As described in the Background, knockdown ASOs/siRNAs can bind to the target transcript and downregulate (pre-)mRNA expression. Knockdown strategies can be utilized in cases where the pathomechanism is a result of overexpression, toxic GoF, or DN effects (Lauffer, van Roon-Mom, Aartsma-Rus, & N = 1 Collaborative, 2024). This section covers the following topics:

1. Considerations for pathomechanism
2. Considerations for dosage sensitivity
3. Important considerations for different inheritance patterns
4. Strategies for searching the literature for gapmer ASOs and siRNAs

Considerations for Pathomechanisms

Variants to be considered for knockdown approaches using gapmer ASOs or siRNA are toxic GoF and DN variants and copy number gains. Strategies on how to assess a variant mechanism are discussed in Step 2. Please ensure you have sufficient evidence of GoF or DN pathomechanism before proceeding further.

Considerations for Dosage Sensitivity

Before proceeding with the development of a knockdown ASO, it is crucial to consider dosage sensitivity. Ideally, a knockdown strategy would be employed when loss-of-function is not expected to cause disease. However, such cases are rare. More commonly, alterations in gene dosage are an underlying cause of disease and must be considered when developing knockdown ASOs. Here it becomes important to be aware of the different inheritance patterns that are implicated in a gene.

If complete loss-of-function is not tolerated[14], but the loss of one gene copy is, knockdowns, such as allele-selective silencing of the mutant allele, can still be considered. Resources such as PubMed, pLI scores and LOEUF scores (both available via gnomAD), DECIPHER dosage sensitivity track (using pHaplo, Collins et al., 2022), and ClinGen dosage sensitivity score can be used to determine this. We consider a curation from a reputable, independent source, such as the ClinGen consortium, the highest level of evidence for dosage sensitivity.

An indication of whether the loss of one allele is tolerated can also be gained from population databases such as gnomAD; if there are carriers of LoF variants in the general population, it can be assumed that the loss of one allele is safe. The same considerations apply to carriers of homozygous LoF variants within a gene that implies that the complete loss of this gene is

[14] Intolerance to complete loss-of-function can, for example, be deducted from the fact that an autosomal recessive disorder is associated with the gene. Note, if the AR disorder is much milder than the AD disorder, one could still consider a knockdown approach.

tolerated. Typically, if the loss of one gene copy is not a known cause of disease and has been observed in healthy cohorts, knockdown ASOs can be considered.

Another factor to consider is haploinsufficiency. As described in Step 2, haploinsufficiency refers to a situation in which one healthy, wildtype allele does not generate sufficient protein product to preserve the physiological state (Deutschbauer et al., 2005). One can utilize resources such as gnomAD's pLI and LOEUF scores, pHaplo score, and ClinGen's dosage sensitivity score (preferred) to determine whether haploinsufficiency is a cause of disease. Typically, a pLI score of equal to or greater than 0.9, the top three deciles of LOEUF scores, a pHaplo score of 0.86 or greater, or a ClinGen haploinsufficiency score of 3 (sufficient evidence) indicates haploinsufficiency being a cause of disease (please note that this should be verified by functional evidence available in the literature). In cases where haploinsufficiency is a known cause of disease, a knockdown approach should only be considered if the disease caused by haploinsufficiency is less severe than the disorder associated with the GoF or DN variants. Though phenotype considerations are beyond the scope of this guideline, one can consider using OMIM, PubMed, and GeneReviews to assess the genotype-phenotype relation.

One should also consider whether hypomorphic alleles are a cause of disease. Hypomorphic alleles are alleles which show partial loss-of-function (sometimes referred to as "leaky" alleles because there is some retention of protein function). In such cases, it is important to consider associated phenotypes, especially if partial loss-of-function is disease causing.

For diseases that are tolerant to complete loss of function (knockout), targeting both alleles may be tolerated.

The use of an allele-selective ASO targeting a SNP or the variant site should always be considered for DN variants. DN variants impact the wildtype product and therefore may result in a functional product loss of greater than 50%. Hence, it is critical that an ASO is designed to specifically target the DN allele while keeping the wildtype product intact to recapitulate as much function as possible. Though this would greatly limit the ASO design, allele-selective ASOs in these scenarios would help ensure that at least 50% of the wildtype function remains (by targeting only the mutant allele). Of course, considerations regarding the severity of a potential haploinsufficiency phenotype remain.

Important considerations for different inheritance patterns

In the case of X-linked and Y-linked disorders, further considerations are necessary. For males, downregulation of a gene on the X or Y chromosome can lead to a complete loss of the gene function and before embarking on such an approach it should be known that this complete loss is tolerated. For X-linked disorders in females, it should be considered how X inactivation (discussed in Step 1) will affect gene dosage and whether a knockdown approach is safe. When considering a knockdown ASO, it is necessary to know whether the gene of interest escapes XCI or might reflect skewed XCI. Each of these situations will result in a different gene dosage, and therefore a different level of maximum expression to reduce.

One X-linked case example is *CLCN4*-related condition, an XLD disease caused by a single pathogenic variant in the *CLCN4* gene (Palmer, E. E. et al., 2018). It affects males and females differently depending on the pathomechanism (Palmer, Elizabeth Emma et al., 1993). LoF variants do no harm in females, but cause the disease in males. On the other hand, DN and GoF variants cause disease in males and females. An allele-selective knockdown ASO would therefore be required for DN and GoF variants in this gene for females which would allow the continued expression of the healthy copy in cells where X inactivation inactivates the mutant copy

ASO Check

In addition to the recommended assessment strategies, assessors should review the literature for knockdown ASO/siRNA strategies. This review can be performed either as the final step to validate the assessment strategy or earlier in the assessment process (see Step 0). Specifically, for knockdown ASOs, it is crucial to evaluate whether a knockdown approach has been developed for other DN or GoF variants in the same gene, and that this approach has been validated at the RNA and protein level and shown to rescue the phenotype at the functional level (pre-clinical work is sufficient). Take note if an allele-selective approach was used, this may limit ASO design and affect outcomes. Further, an ASO might have been developed that is specific for a variant (other than the one under assessment) or a SNP. Information on phasing of an individual's variant with that SNP will then have to be obtained. The gene is still "eligible" for knockdown according to the criteria outlined in Table 6 even if the already available ASO is not suitable for the individual under assessment. This means a new ASO would have to be designed and developed for that individual.

In cases of conflicting evidence regarding ASO developments and the feasibility of an ASO therapy, consider the quality of the research, the types of experiments conducted, the nature of the results shared, and the publication date. The evaluation of literature on existing ASOs should be carried out at the assessor's discretion, with a critical and discerning approach.

To help with identifying available ASOs/siRNA for knockdown approaches, we recommend a search term in Pubmed like this, text in bold would need to be adjusted for the gene and variant being examined:

SCN2A AND ((ASO) OR (AON) OR (antisense oligonucleotide) OR (AOs) OR (siRNA) OR (RNAi) OR (gapmer) or (knockdown))

SCN2A AND ((ASO) OR (AON) OR (antisense oligonucleotide) OR (AOs) OR (siRNA) OR (RNAi)) AND ((***p.R853Q***) OR (**p.Arg853Gln**) OR (***c.2558G>A***))

Once all this information is collected, one can use Table 6 to classify the variant's eligibility towards ASO knockdowns as "eligible", "likely", "unlikely", or "not eligible".
Variants, where not enough evidence exists on the pathomechanism or the dosage sensitivity cannot be assessed until more evidence is collected.

Table 6: Classification of variants for their eligibility towards knockdown

Classification	Criteria
Eligible	ASO/RNAi/siRNA has already been developed and shown to work with available functional evidence (i.e., evidence of knockdown rescuing function, pre-clinical data is sufficient)

Likely eligible	Variant is Gain-of-Function or Dominant-Negative (functionally proven)
	
	AND
	
	{
	Gene is tolerant to the reduction of gene dosage (i.e., gene is NOT haploinsufficient)
	
	AND/OR
	
	Individuals with heterozygous LoF variants are present in population databases/described in medical literature, such that high penetrance for probably severe disease phenotypes are unlikely
	}

Unlikely eligible	Variant is Gain-of-Function or Dominant-Negative (functionally proven) **BUT** Heterozygous LoF/Haploinsufficiency/Hypomorphic variants has/have been associated with a disease **OR** X-linked gene that undergoes X-chromosome inactivation
Not eligible	Intolerant to reduction (i.e., gene dosage is tightly regulated in humans, and knockdown is expected to lead to serious phenotypic consequences) **OR** Evidence that an ASO cannot be developed, shown by two independent investigations at the protein/functional level or one investigation with a convincing explanation why an ASO cannot be developed.

Section C - Considerations for Upregulation from the Wildtype Allele

For disorders caused by haploinsufficiency, one functional wildtype gene copy remains. In these situations, one can use ASOs to upregulate the wildtype allele. This approach, also known as targeted augmentation of nuclear gene output (TANGO) (Lim et al., 2020), can include skipping of poison exons, downregulating naturally occurring antisense transcripts, and targeting UTR regulator elements such as upstream open reading frames, all of which can increase the gene product. For a more detailed explanation of these approaches, see the Background, Fig. 3, Fig. 11, and the following publications:

1. Lim et al. (2020) https://doi.org/10.1038/s41467-020-17093-9
2. Mittal et al. (2022) https://doi.org/10.1016/j.medj.2022.08.006
3. Felker et al. (2023) https://doi.org/10.1016/j.gim.2023.100884

For convenience, the data found in the supplementary tables and databases from the three aforementioned publications have also been compiled into one excel sheet (most recent excel sheet can be found on the N1C website). Assessors can use this sheet to look up poison exons, naturally occurring antisense transcripts, and upstream open reading frames found in each of these publications. It is highly encouraged that assessors do not rely on this one resource alone, but also utilize it in conjunction with the other strategies discussed below. Throughout this section, we will refer to tools and databases that can be utilized to better supplement and verify the findings from the three papers.

Please note that, unlike the other ASO strategies, these guidelines will not provide details on how to classify variants as “likely”, “unlikely", or “not eligible”. The upregulation of wildtype transcripts through the aforementioned strategies heavily depends on the availability of functional evidence for key regulatory elements. These strategies are not comparable to one another, and different approaches must be employed depending on the availability of regulatory elements. Instead, readers can reference this section to learn about the different strategies and resources they can utilize for their own analyses. However, in the case where a wildtype upregulation approach has already been established for a given gene with sufficient functional evidence, we consider a variant as “eligible” for wildtype upregulation. The sole existence of a poison exon, NAT, or other information in the papers and our combined table without further data does not qualify as sufficient evidence and does not support the classification “eligible” for wildtype upregulation.

This section covers:

1. Targeting naturally occurring antisense transcripts
2. Targeting upstream open reading frames
3. Targeting poison exons and non-productive alternative splicing events

Naturally Occurring Antisense Transcripts

Through downregulating naturally occurring antisense transcripts via gapmer ASOs, it is - in some instances - possible to upregulate gene expression (Fig. 3, and Example 9 discussing upregulation of *UBE3A* (Dindot et al., 2023)). Assessors can utilize resources such as PubMed, UCSC Genome Browser, HUGO Gene Nomenclature Committee, and Ensembl to search for known antisense transcripts for a given gene. If searching through a genome browser like UCSC, naturally occurring antisense transcripts would be found to run in the reverse direction on the complement strand, and would be noted as a non-coding RNA. It is important to verify that this transcript is indeed a non-coding antisense transcript, and not another protein coding gene. Not all complementary transcripts are NATs. To verify, we suggest utilizing databases such as HUGO and paying attention to alias names or symbols that may indicate its function as a non-coding antisense transcript (i.e. “AS”, “antisense”, etc.). Please also note that antisense transcripts can be located in different positions in respect to the sense transcript, which will influence whether they are good targets. To learn more about antisense transcripts and their directions, we recommend to study Werner et al. 2024 and Zinad et al 2017.

Assessors should also consider the level at which these antisense transcripts are expressed in the tissue of interest. Databases such as GTEx and the Human Protein Atlas can assist with this. No general threshold of expression level can be provided, as this is gene dependent. However, assessors should pay attention to which tissues the transcript is enriched in, and how expression levels compare to the gene we wish to upregulate. Note that the existence of an antisense transcript alone is not enough to consider an ASO development. Factors such as tissue expression, regulatory function, and orientations of the antisense transcripts (see above) must be well understood before considering this approach. For this reason, we highly encourage a literature search through PubMed to supplement the above assessment. Functional evidence must be available, describing the role of the transcript as a downregulator of the gene of interest. Search terms can include the name of the gene of interest, followed by the term “antisense transcript”, or the official name of the antisense transcript or host gene itself.

Upstream Open Reading Frames

Targeting uORFs for upregulation of the WT allele can be approached in two different ways. One option is that an ASO can be designed to target and block the regulatory elements in untranslated regions, including uORFs (Fig. 3). Resources such as PubMed can be used to check whether there is a uORF that can be targeted in the relevant transcript. Additionally, the supplemental table 2 in the paper by Mittal et al. (2022) indicates for which genes uORFs have been identified. Note that not all uORFs act through an inhibitory mechanism, and proper characterization of their mechanism is essential in determining their eligibility as ASO targets. To collect this information, we recommend a review of the literature and available functional studies. The uORF should also be present in relevant and highly expressed transcripts in the target tissue.

In addition to blocking uORFs (as described in Fig. 3), another approach is skipping an exon in the UTR that contains a uORF start site (Fig. 11). This strategy has been explored in articles by Feng et al. (2025) and Beer Wells et al. (2025). The Feng et al. study identified several transcript isoforms of *BDNF* regulated by uORFs. Experimental validation confirmed uORF-mediated

regulation in five *BDNF* transcripts. One transcript showed marked upregulation when an exon was removed from its 5′ UTR. Therefore, if considering this approach, one must assess the following:

- Expression levels of the alternate transcript regulated by the uORF (verify that it is expressed in the tissue of interest at sufficient levels, tools such as GTEx can be used for this)
- uORFs are experimentally validated
- A “skippable exon” in the context of pORF upregulation is identified, defined as 5’ UTR exons containing a minimum of one uORF start codon that does **not** contain the translation start site

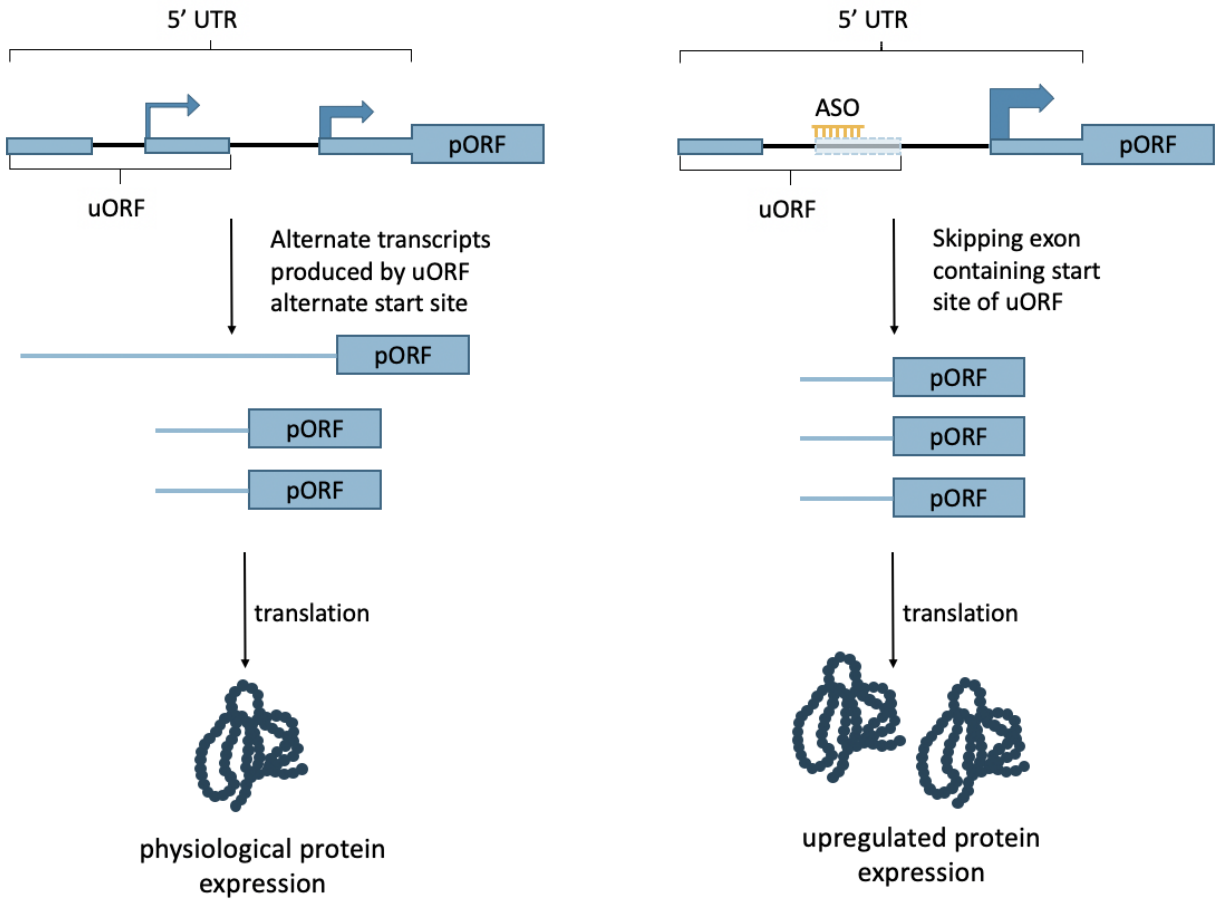


Figure 11: Skipping of untranslated exons to promote translation of pORF

ASO can be used to skip untranslated exons containing start sites of the upstream open reading frame (uORF), promoting translation of the primary open reading frame (pORF), increasing wildtype gene product.

Poison Exons and Non-Productive Alternate Splicing Events

One can design ASOs which target non-productive alternate splicing events including poison exons (Fig. 3). These ASOs would utilize the splice-switching approaches described in the splice-correction and exon skipping sections to promote canonical splicing of the wildtype transcript. Resources such as PubMed, Ensembl, VastDB, and the UCSC Genome Browser can aid in determining alternate splicing events in the transcript of interest. Additionally, supplemental table 2 from Mittal et al. (2022), supplemental data 2 from Lim et al. (2020), and supplemental data 1 from Felker et al. (2023) list all identified poison exons in these papers. Databases such as GTEx can further assist by determining the expression level of alternate transcripts in the target tissue. As noted earlier, no specific threshold is available when reviewing transcript levels, as this will differ by gene and disease considerations. Note that some alternate splicing events are crucial for the production of important transcripts and isoforms. Generally, skipping a poison exon should lead to increased WT levels high enough to alter the disease state. The most prominent example of poison exon skipping in *SCN1A* for Dravet syndrome was possible because up to 80% of the transcripts contain this poison exon (Tang et al., 2025).

For more resources and strategies on upregulating wildtype gene products, please reference the *Useful Tools* section.

Important considerations for different inheritance patterns and pathomechanisms

The considerations outlined here are mainly applicable to diseases associated with haploinsufficiency, which is most likely caused by LoF variants in AD disorders. In rare cases, one could also consider applying upregulation of wildtype allele strategies in X-linked disorders in females where there is sufficient evidence that upregulation from the second X chromosome is possible. Please always consider the challenges caused due to X inactivation.

Rarely, one can also consider WT upregulation for DN variants, with the goal of producing enough wildtype protein product to ameliorate the phenotypes caused by the DN protein. However, this may be made nearly impossible, as it will require the design of an allele-specific upregulation ASO.

Exceptionally, upregulation of transcripts can also be considered for autosomal recessive disorders. In the case of partial LoF variants where an upregulation of the transcript would lead to a small increase in protein function, and where a few percentage increase in function can lead to a meaningful clinical benefit, such an approach can be considered. A good example of diseases where this strategy might be worthwhile considering are metabolic disorders, where enzymatic function can be measured, and small increases lead to meaningful effects.

ASO Check

Also for the upregulation of wildtype allele approaches, one can search whether specific strategies already exist, are under development, or are pursued in clinical trials. Multiple strategies can be utilized to upregulate gene products from the wildtype transcript, and it is therefore important to employ a variety of strategies/approaches in the search terms to ensure a comprehensive review of the field.

While we do not classify the different upregulation approaches, we consider a variant as “eligible” for upregulation if an upregulation strategy has been developed and demonstrated to work with sufficient evidence pre-clinically.

Addendum

Summary “Unable to assess”

A variant is classified “unable to assess” when:

- The variant does not fall into the category of variants for these guidelines, e.g. translocations, variants in non-coding genes, loss of both copies of a gene due to large/whole gene deletions
- The variant description is incorrect
- The inheritance pattern is not known
- The pathomechanism of the variant is not known
- Not enough information on dosage sensitivity is available (for knockdown)
- Intronic variant without sufficient functional information on splicing

Please note that exon skipping strategies are applicable irrespective of inheritance pattern and pathomechanism. Assessors can still review whether the exon is applicable for exon skipping strategies, though this does not exclude the need to understand the pathomechanism or inheritance of the given variant.

Examples

The below examples provide variant assessments for different types of variants. Detailed explanations for all the examples can be found in the accompanying training videos (via YouTube or the N1C website).
To further practice variant assessments, we encourage new assessors to use the test variants in Table S1 and later check the answer keys in File S2 of the original publication.

Table 7: Example variants.

Example assessments of different variant types for which explanatory videos are available via YouTube.

#	Variant and Video Content	Assessment
1	NM_000350.3(ABCA4):c.2626C>T p.(Gln876Ter) [MIM 601691] Corresponding video content: ● Nomenclature check using Mutalyzer ● Searching UCSC to determine exon number ● Searching PubMed for exon skipping ASO ● Recommendations for continued analysis	**Eligibility: Eligible for exon skipping ASO** Explanation: Correct variant description, variant is loss of function in a recessive disorder. Variant is >15 bp downstream of the nearest splice site and is a nonsense variant considered for exon skipping. Variant is located in exon 17/50 in the *ABCA4* gene, exon is a small, in-frame exon. Skipping of in-frame exon 17 has been shown in pre-clinical studies (Kaltak et al., 2023).
2	NM_016589.4(TIMMDC1):c.597-1340A>G [MIM 615534] Corresponding video content: ● Nomenclature check using VariantValidator (and example output of incorrect variant) ● Searching PubMed for splice-switching ASO	**Eligibility: Eligible for splice correction ASO** Explanation: Correct variant description, variant is loss of function in a recessive gene. Variant is deep intronic and >100 bp from the nearest splice site. Variant has been reported multiple times and the effect on splicing has been confirmed via RNAseq (Kremer et al., 2017). Variant causes the insertion of a cryptic exon and a premature stop.

		Pre-clinical data is available that shows evidence of rescue of enzymatic effects upon ASO treatment (Kumar et al., 2022).
3	NM_000533.5(PLP1):c.680dup p.(Cys228LeufsTer5) [MIM 300401] Corresponding video content: ● Checking inheritance patterns using OMIM ● Checking variant mechanism using GeneReviews ● Using UCSC genome browser to determine exon position ● Using ExonViz to determine exon frame	**Eligibility: Not eligible for exon skipping ASO** Explanation: Correct variant description, X-linked recessive gene, frameshift variant leading to an early stop can only be considered for exon skipping (Section A). Variant >15 bp upstream of nearest splice site. Exon 5/7 is out-of-frame and thus not eligible.
4	NM_003793.4(CTSF):c.213+1G>C [MIM 603539] Corresponding video content: ● Checking inheritance patterns using OMIM ● Checking variant mechanism using ClinVar ● Assessing splicing effects of variants through a literature search	**Eligibility: Not eligible for splice correction ASO** Explanation: Correct variant description, autosomal recessive gene. Variant is a canonical splice site variant and was shown to lead to skipping of exon 1, leading to disease (Di Fabio et al., 2014). Variant is within 5 bp of canonical splice site.
5	NM_000277.3(PAH):c.611A>G p.(Tyr204Cys) [MIM 612349] Corresponding video content: ● Checking inheritance patterns using OMIM ● Checking variant mechanisms using OMIM and a literature search ● Assessing splicing effects of variants through a literature search	**Eligibility: Unlikely for splice correction ASO** Explanation: Correct variant description, variant is in a recessive gene, variant is a missense variant with evidence of effect on splicing. Variant affects mRNA splicing and results in a 32-amino acid deletion of the PAH enzyme (Ellingsen, Knappskog, & Eiken, 1997).

	● Assessing effects of missense variant through a literature search ● Checking the position of the variant respective to canonical splice sites using UCSC	Variant is classified as unlikely since Ellingsen et al. could only identify a small change in enzymatic activity when generating the enzyme with the missense variant.
6	NM_025152.3(NUBPL):c.815-27T>C [MIM 613621] Corresponding video content: ● Checking inheritance patterns using OMIM ● Assessing variant mechanisms and splicing effects through a literature search	**Eligibility: Unlikely eligible for splice correction ASO** Explanation: Correct variant description. Variant associated with autosomal recessive disorder. Variant is intronic with evidence of effect on splicing. Variant is not within 5bp of the canonical splice site, but in the 5 to 100 bp region. Variant affects a branchpoint. The branchpoint is weakened but not destroyed (30% of wildtype transcript remains) (Maclean, Kimonis, & Balk, 2018). Therefore, this variant is unlikely eligible for splice correction ASO.
7	NM_024312.5(GNPTAB):c.3503_3504 del p.(Leu1168fs) [MIM 607840] Corresponding video content: ● Searching the literature for an already existing ASO ● Checking inheritance patterns using OMIM ● Assessing variant mechanisms and eligibility for exon-skipping using the literature	**Eligibility: Unlikely eligible for exon skipping ASO** Explanation: Correct variant description. ASO exists but has only been validated at the RNA level (not the protein level). Variant is associated with autosomal recessive disorder and is loss-of-function. Variant is in exon 19/21. Exon is in-frame and codes for 4.5% of the coding transcript. A stop codon does not form when exon is skipped. Exon codes for a stealth domain, but does not meet any of the exclusion criteria in Table 3 (Matos et al., 2020). Therefore, this variant is unlikely eligible for an exon skipping ASO.

8	NM_001040142.2(SCN2A):c.5645G>A p.(R1882Q) [MIM 182390] Corresponding video content: ● Nomenclature check using variant validator ● Inheritance pattern check using OMIM ● Pathomechanism check using ClinVar and by conducting a literature search ● Dosage sensitivity assessment using ClinGen, pLI, and the literature ● Conducting an ASO check	**Eligibility: Eligible for knockdown** Explanation: Correct variant description. ASO exists and has been validated clinically. Variant is associated with an autosomal dominant gain-of-function disorder. Haploinsufficiency and loss-of-function is also a known cause of disease, but is arguably associated with less severe phenotypes. Variant classified as “eligible” because preclinical evidence of knockdown approach exists.
9	NM_001165963.4(SCN1A):c.3733C>T p.(R1245Ter) [MIM 182389] Corresponding video content: ● Nomenclature check using variant validator ● Inheritance pattern check using OMIM ● Pathomechanism check using ClinVar ● Dosage sensitivity assessment using ClinGen and pLI ● Checking for upregulation (i.e. TANGO) methods using the literature ● Conducting an ASO check ● Brief discussion on other upregulation of wildtype allele approaches using NM_130839.5 (UBE3A):c.67C>T, p.(Arg23*) as an example [MIM *601623]	**Eligibility: Eligible for upregulation of wildtype allele** Explanation: Correct variant description. ASO exists and has been clinically validated. Variant is associated with autosomal dominant loss-of-function disorder. Haploinsufficiency and loss-of-function is a known cause of disease. Variant is found in an exon not eligible for canonical exon skipping. Poison exons identified in the literature, and have been clinically validated.

10	NM_003793.4(CTSF):c.264del p.(Cys89fs) [MIM 603539] Corresponding video content: ● Checking inheritance patterns using OMIM ● Checking variant mechanism using ClinVar ● Using UCSC genome browser to determine exon position ● Using ExonViz and Ensembl to determine exon frame ● Using RefSeq and UniProt to determine exon size ● Using UniProt to assess corresponding protein function ● Recommendations for continued analysis	**Eligibility: Likely eligible for exon skipping ASO** Explanation: Correct variant description, autosomal recessive gene, variant is a frameshift variant, thus can only be assessed for exon skipping (Section A). Exon 2/13 is in-frame, approx. 7% of the protein coding region. Skipping of exon does not create a stop codon. No functional domain is known and no mutational hotspot identified.
11	NM_000170.3(GLDC):c.538C>T p.(Gln285Ter) [MIM 238300] Corresponding video content: ● Inheritance pattern check using OMIM ● Pathomechanism check using ClinVar ● Using UCSC genome browser to determine exon position and frame ● Using UCSC to check whether exon skipping forms a stop codon	**Eligibility: Not eligible for exon skipping** Explanation: Variant description is correct. Variant is associated with autosomal recessive loss-of-function disorder. Variant is not in the first or last codon exon. Variant is not in an out-of-frame exon. However, variant is found in an exon which will form a stop codon if skipped, and the two neighboring exons join together. Therefore, this variant is not eligible towards canonical exon skipping ASOs.
12	Unable to Assess Examples Corresponding video content: ● Example of a mitochondrial variant ● Example of incorrect variant notation ● Example of a missense variant with an unknown pathomechanism	**Eligibility: Unable to assess** All variants discussed in the video are ineligible for assessment either because they are unable to be assessed by these specific guidelines, or not enough information is available to proceed with assessment.

Table 8: Additional Example Variants and their Assessments.

New examples include cases addressing the newly added sections for v2.0 of the guidelines.

#	Variant and Video Content	Assessment
13	NM_000441.2(*SLC26A4*):c.919-2A>G	**Eligibility: Eligible for exon inclusion ASO** Explanation: Correct variant description. Variant associated with autosomal recessive disorder. Variant is intronic with evidence of effect on splicing. Variant is within 5 bp of the canonical splice site. Variant affects acceptor site, but some canonical transcript still present (Lee et al., 2014). Pre-clinical data available for exon inclusion in patient-derived cells (Feng et al., 2022)
14	NM_001130438.3(*SPTAN1*):c.6619_6621del	**Eligibility: Eligible for knockdown ASO** Explanation: Correct variant description. Variant associated with autosomal dominant disorder. Variant is associated with dominant-negative mechanism (Saitsu et al., 2010), and therefore, **an allele-selective approach must be considered**. Haploinsufficiency of *SPTAN1* is also disease causing, but an allele-selective ASO is shown to work pre-clinically (Wang et al., 2025).
15	NM_000828.5(GRIA3):c.1961C>T	**Eligibility: Unlikely eligible for knockdown ASO** Explanation: Correct variant description. Variant associated with X-linked disorder. Variant is associated with gain-of-function mechanism (Rinaldi et al., 2023).

		Loss-of-function of *GRIA3* is also disease causing, and X inactivation is possible (according to ClinGen), therefore knockdown may not be a suitable approach.
16	NM_021008.4(DEAF1):c.670C>T	**Eligibility: Likely eligible for knockdown ASO** Explanation: Correct variant description. Variant associated with autosomal dominant inheritance (some variants in these gene can be autosomal recessive, this variant was found inherited in an autosomal dominant manner). Variant is associated with dominant-negative mechanism (Vulto-van Silfhout et al., 2014). Haploinsufficiency is not thought to be disease-causing but a full knockdown is not possible as an autosomal recessive disorder is also associated with the *DEAF1* gene. Thus, given that the variant is dominant-negative, the variant is likely eligible for a knockdown approach using an allele-selective ASO.

Useful Tools

This is a collection of tools that can be useful during variant assessment. Most of them are already listed throughout the text. Tools are listed matching the different steps of these guidelines.

Step 0 - Variant check
HGVS nomenclature:https://varnomen.hgvs.org/
Mutalyzer: https://mutalyzer.nl/ (cannot do deep intronic variants in c. notation but works with g. notation)
VariantValidator: https://variantvalidator.org/

Step 1 - Assessment of pattern of inheritance and disease type
Gene Cards: http://www.genecards.org/
OMIM: https://omim.org/
Gene Reviews: https://www.ncbi.nlm.nih.gov/books/NBK1116/
Orphanet: https://www.orpha.net/en/disease
Monarch Initiative: https://monarchinitiative.org
Gene Curation Coalition: https://thegencc.org/
ClinGen: https://www.clinicalgenome.org/
gnomAD: https://gnomad.broadinstitute.org/ (population frequency of a variant)
DisGeNet: https://www.disgenet.org/
Gene2Phenotype: https://www.ebi.ac.uk/gene2phenotype

Step 2 - Assessment of pathomechanism of genetic variant
HGMD: https://www.hgmd.cf.ac.uk/
DECIPHER: www.deciphergenomics.org/
ClinVar: https://www.ncbi.nlm.nih.gov/clinvar/
Mastermind Genomenon: https://mastermind.genomenon.com/
Pubmed: https://pubmed.ncbi.nlm.nih.gov/
Franklin: https://franklin.genoox.com/clinical-db/home
LOVD: https://www.lovd.nl/
Varsome: https://varsome.com/

Step 3 - Evaluation of splicing effects
HGMD: https://www.hgmd.cf.ac.uk/
DECIPHER: https://www.deciphergenomics.org/
ClinVar: https://www.ncbi.nlm.nih.gov/clinvar/
Mastermind Genomenon: https://mastermind.genomenon.com/
Pubmed: https://pubmed.ncbi.nlm.nih.gov/

Search engine → web-search the variant

Section A - Considerations for canonical exon skipping

UCSC Genome Browser:https://genome.ucsc.edu/
ExonViz:https://exonviz.rnatherapy.nl/
Ensembl: https://www.ensembl.org/index.html
Metadome: https://stuart.radboudumc.nl/metadome/
UniProt: https://www.uniprot.org/
Protein Database: https://www.rcsb.org/
Alphafold: https://alphafold.ebi.ac.uk/
ES-NDD: https://es-ndd.broadinstitute.org/
Franklin: https://franklin.genoox.com/clinical-db/home
ExonSkip DB: https://ccsm.uth.edu/ExonSkipDB/
VulExmMap:https://vulexmap.compbio.sdu.dk/
Genetic Therapy Generator Toolkit (GTGT):https://gtgt.rnatherapy.nl/

Section B - Considerations for Downregulation
gnomAD for pLI and Loef: https://gnomad.broadinstitute.org/
Dosage sensitivity map:UCSC Genome Browser: News Archives
ClinGen dosage sensitivity curation: https://search.clinicalgenome.org/kb/gene-dosage?page=1&size=25&search=
DECIPHER dosage sensitivity track:
https://genome.ucsc.edu/goldenPath/newsarch.html#022124

Section C - Considerations for upregulation from the wildtype allele
Antisense transcripts via HUGO: Search results | HUGO Gene Nomenclature Committee (genenames.org)
Ribo uORF:https://rnainformatics.org.cn/RiboUORF/
GTEx: https://gtexportal.org/home/transcriptPage
VastDB: https://vastdb.crg.eu/wiki/Main_Page

Further tools
MobiDetails: https://mobidetails.iurc.montp.inserm.fr/MD/
ProteinPaint: https://proteinpaint.stjude.org/
Proteinatlas:http://www.proteinatlas.org/
FLIBase - Full length isoforms in cancers and normal tissues: http://flibase.org/#/home

Abbreviations

AD autosomal dominant

AR autosomal recessive

ASO Antisense Oligonucleotide

CNV company number variant

DN dominant negative

GoF gain of function

HGVS Human Genome Variation Society

LoF loss of function

NMD nonsense mediated decay

ORF open reading frame

pORF primary open reading frame

SE splice enhancer

siRNA small interfering RNA

SNP Single nucleotide polymorphism

SS splice silencer

TANGO targeted augmentation of nuclear gene output

uORF upstream open reading frame

UTR untranslated region

WT wildtype

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